Fundamentals and exchange rate forecastability with simple machine learning methods☆

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Appendix A. Detailed data description

We recall that, as indicated in Section 4, the data used is freely available for download (Amat et al., 2018a).

The original fundamentals dataset of Molodtsova and Papell (2009) was extended to the period 1973–2014 whenever possible. For the discontinued time-series we used instead Datastream as the principal source. For those, as there were typically more time-series available without a clear advantage of one over the others, only those that were finally used to generate predictions presented in the paper are listed (we did not get qualitatively different results using the others). Data was obtained from Datastream on 30/01/2014. When the code from the original data series was known, it was given instead. All series are monthly unless noted.

Quarterly time-series (there were only three of them) were transformed into monthly ones by local quadratic interpolation, i.e., a local quadratic polynomial was fit for each set of three adjacent quarterly observations; then, the monthly observations right before and after the central quarterly observation (the second one in the set) were imputed using the values of the quadratic polynomial. The output gap was estimated for each country with at least 24 data points. For the Hodrick-Prescott filter we used $\lambda = 129\,600$ as advocated by Ravn and Uhlig (2002).

The exact data series (and the periods for which they could be used) are described in detail in Tables A.8–A.9 below. In particular, the three aforementioned quarterly time-series are indicated therein by a "quarterly" comment in their descriptions.

Real-time data on CPI, money stocks and industrial production for the period February 1999 to April 2017 were obtained from the OECD MEI data base (`http://stats.oecd.org/mei/default.asp?rev=1`) for countries whose currencies were not superseded by the euro (i.e., the British pound, Japanese yen, Swiss franc, Canadian dollar, Swedish krona, Danish kronor, Australian dollar and the comparison currency – the U.S. dollar). The time-series for M2 U.S. money supply until March 2001 (as there is a break in OECD reporting of M3/M2 series) were directly taken from the Federal Reserve historical releases website (`https://www.federalreserve.gov/releases/h6/`).

Table A.8: Data series description, part I.

Description	Series name	Intermediate source (if any)	Original source	Earliest date after March 1973 when the series is available	End date
End-of-month nominal exchange rate, U.S.-U.K	UKI..AG.	Datastream	IFS	Mar-73	Dec-14
End-of-month nominal exchange rate, Japan -- U.S.	JPI..AE.	Datastream	IFS	Mar-73	Dec-14
End-of-month nominal exchange rate, Switzerland -- U.S.	SWI..AE.	Datastream	IFS	Mar-73	Dec-14
End-of-month nominal exchange rate, Canada -- U.S.	CNI..AE.	Datastream	IFS	Mar-73	Dec-14
End-of-month nominal exchange rate, Sweden -- U.S.	SDI..AE.	Datastream	IFS	Mar-73	Dec-14
End-of-month nominal exchange rate, Denmark -- U.S.	DKI..AE.	Datastream	IFS	Mar-73	Dec-14
End-of-month nominal exchange rate, U.S.-- Australia	AUI..AG.	Datastream	IFS	Mar-73	Dec-14
End-of-month nominal exchange rate, France -- U.S.	FRI..AE.	Datastream	IFS	Mar-73	Dec-14
End-of-month nominal exchange rate, Germany -- U.S.	BDI..AE.	Datastream	IFS	Mar-73	Dec-14
End-of-month nominal exchange rate, Italy -- U.S.	ITI..AE.	Datastream	IFS	Mar-73	Dec-14
End-of-month nominal exchange rate, Netherlands -- U.S.	NLI..AE.	Datastream	IFS	Mar-73	Dec-14
End-of-month nominal exchange rate, Portugal -- U.S.	PTI..AE.	Datastream	IFS	Mar-73	Dec-14
M1 money supply, n.s.a., U.S.	USI59MA.A	Datastream	IFS	Mar-73	Nov-14
Notes and coins in circulation outside the Bank of England, n.s.a., U.K.	UKAVAA..	Datastream	Bank of England	Mar-73	Dec-14
M1 money supply, billions of yens, n.s.a, Japan	JPI59MA.A	Datastream	IFS	Mar-73	Nov-14
Narrow money, billions of Swiss franks, n.s.a., Switzerland	SWI34...A	Datastream	IFS	Mar-73	Oct-14
M1 money supply, billions of Canadian dollars, n.s.a., Canada	CNI59MADA	Datastream	IFS	Mar-73	Oct-14
M0 money supply, millions of Swedish kronors, n.s.a., Sweden	SDM0....A	Datastream	Statistics Sweden	Mar-73	Dec-14
M1 money supply, millions of Danish kroners, n.s.a., Denmark	DKOMA027A	Datastream	MEI, OECD	Mar-73	Feb-14
M1 money supply, millions of Australian dollars, n.s.a., Australia	AUOMA027A	Datastream	MEI, OECD	Mar-73	Feb-14
M1 money supply, billions of French franks, n.s.a., France	13259MA.ZF..	Molodtsova and Papell (2009)	IFS	Dec-77	Dec-98
M1 money supply, billions of Deutsche marks, s.a., Germany	13459MACZF...	Molodtsova and Papell (2009)	IFS	Mar-73	Dec-98
M2 money supply, billions of Italian liras, n.s.a., Italy	13659MB.ZF...	Molodtsova and Papell (2009)	IFS	Dec-74	Dec-98
M2 money supply, billions of Dutch guilders, n.s.a., Netherlands	13859MB.ZF...	Molodtsova and Papell (2009)	IFS	Mar-73	Dec-98
M1 money supply, billions of Portuguese escudos, n.s.a., Portugal	18259MA.ZF...	Molodtsova and Papell (2009)	IFS	Dec-79	Dec-98
Federal Funds Rate, United States	USI60B..	Datastream	IFS	Mar-73	Dec-14
Money Market Rate, U.K.	UKI60B..	Datastream	IFS	Mar-73	Nov-14
Money Market Rate, Japan	JPI60B..	Datastream	IFS	Mar-73	Dec-14
3-Month Euro Deposits, Switzerland	SWOIR075R	Datastream	MEI, OECD	Jan-74	Nov-14
Treasury Bills rate, Canada	CNI60C..	Datastream	IFS	Mar-73	Dec-14
Money Market Rate, Sweden	SDI60B..	Datastream	IFS	Mar-73	Oct-14
Money Market Rate, Denmark	DKI60B..	Datastream	IFS	Mar-73	Dec-14
Money Market Rate, Australia	AUI60B..	Datastream	IFS	Mar-73	Dec-14
Money Market Rate, France	13260B..ZF...	Molodtsova and Papell (2009)	IFS	Mar-73	Dec-98
Money Market Rate, Germany	13460B..ZF...	Molodtsova and Papell (2009)	IFS	Mar-73	Dec-98
Money Market Rate, Italy	13660B..ZF...	Molodtsova and Papell (2009)	IFS	Mar-73	Dec-98
Money Market Rate, Netherlands	13860B..ZF..	Molodtsova and Papell (2009)	IFS	Mar-73	Dec-98
Money Market Rate, Portugal	18260B..ZF..	Molodtsova and Papell (2009)	IFS	Jan-83	Dec-98

Table A.9: Data series description, part II.

Description	Series name	Intermediate source (if any)	Original source	Earliest date after March 1973 when the series is available	End date
Industrial production, s.a., U.S.	USI66..CE	Datastream	IFS	Mar-73	Dec-14
Industrial production, s.a., U.K.	UKI66..CE	Datastream	IFS	Mar-73	Nov-14
Industrial production, s.a., Japan	JPI66..CE	Datastream	IFS	Mar-73	Oct-14
Industrial production, s.a., Switzerland, quarterly	SWQ66..BH	Datastream	IFS		
Industrial production, s.a., Canada	CNI66..CE	Datastream	IFS	Mar-73	Oct-14
Industrial production excluding construction, s.a., Sweden	SDOPRI35G	Datastream	MEI, OECD	Mar-73	Oct-14
Industrial production, s.a., Denmark	DKI66..BH	Datastream	IFS	Jan-74	Oct-14
Industrial production, s.a., Australia, quarterly	AUQ66..CE	Datastream	IFS		
Industrial production, s.a., France	13266..CZF...	Molodtsova and Papell (2009)	IFS	Mar-73	Dec-98
Industrial production, s.a., Germany	13466..CZF...	Molodtsova and Papell (2009)	IFS	Mar-73	Dec-98
Industrial production, s.a., Italy	13666..CZF...	Molodtsova and Papell (2009)	IFS	Mar-73	Dec-98
Industrial production, s.a., Netherlands	13866..CZF...	Molodtsova and Papell (2009)	IFS	Mar-73	Dec-98
Industrial production, s.a., Portugal	18266..BZF...	Molodtsova and Papell (2009)	IFS	Mar-73	Dec-98
Consumer price index, U.S.	USI64...F	Datastream	IFS	Mar-73	Dec-14
Consumer price index (retail price index), U.K.	UKI64B..F	Datastream	IFS	Mar-73	Nov-14
Consumer price index, Japan	JPI64...F	Datastream	IFS	Mar-73	Nov-14
Consumer price index, Switzerland	SWI64...F	Datastream	IFS	Mar-73	Dec-14
Consumer price index, Canada	CNI64...F	Datastream	IFS	Mar-73	Nov-14
Consumer price index, Sweden	SDI64...F	Datastream	IFS	Mar-73	Dec-14
Consumer price index, Denmark	DKI64...F	Datastream	IFS	Mar-73	Dec-14
Consumer price index, quarterly, Australia	AUI64...F	Datastream	IFS	Mar-73	Aug-14
Consumer price index, France	13264...ZF...	Molodtsova and Papell (2009)	IFS	Mar-73	Dec-98
Consumer price index, Germany	Consumer prices: all items, 2000=100	Molodtsova and Papell (2009)	MEI, OECD	Mar-73	Dec-98
Consumer price index, Italy	13664...ZF...	Molodtsova and Papell (2009)	IFS	Mar-73	Dec-98
Consumer price index, Netherlands	13864...ZF...	Molodtsova and Papell (2009)	IFS	Mar-73	Dec-98
Consumer price index, Portugal	18264...ZF...	Molodtsova and Papell (2009)	IFS	Mar-73	Dec-98

Appendix B. Assessing forecast quality

We denote by T the total number of monthly values to be forecast, from months 1 to T. As is standard in the literature (see Molodtsova and Papell, 2009 and Rossi, 2013) and for essentially the same reasons, namely, the consideration of rolling OLS regressions, we allow for a training period of length $t_0 = 120$ months (10 years) and only evaluate the accuracy of the forecasts in months $t_0 + 1 = 121$ to T. To that end, we consider the root mean square error,

$$\begin{aligned}\text{RMSE} &= \sqrt{\frac{1}{T - t_0} \sum_{t=t_0+1}^{T} \big(\widehat{s}_t - s_t\big)^2} \qquad \text{(B.1)} \\ &= \sqrt{\frac{1}{T - t_0} \sum_{t=t_0+1}^{T} \Big(\big(\widehat{s}_t - s_{t-1}\big) - \big(s_t - s_{t-1}\big)\Big)^2},\end{aligned}$$

and note that (by subtracting and adding the pivotal values s_{t-1}) this root mean square error is for both the logarithms of exchange rates s_t and for changes in logarithms of exchange rates $s_t - s_{t-1}$.

We want to investigate whether the improvements in RMSE of one method over another are statistically significant. Denote by $\widehat{s}'_t$ and $\widehat{s}_t$ the respective forecasts of two methods of interest. We aim to test the hypothesis H_0 that the difference in forecasting accuracy is not significant against the alternative hypothesis H_1 that the second method – the one outputting the forecasts $\widehat{s}_t$ – is significantly better, on average. To that end, the standard practice (see the two tests presented below) is to consider the instantaneous differences in accuracy

$$d_t = \big(\widehat{s}'_t - s_t\big)^2 - \big(\widehat{s}_t - s_t\big)^2 .$$

We denote by

$$\overline{d}_T = \frac{1}{T - t_0} \sum_{t=t_0+1}^{T} d_t \qquad \text{and} \qquad \overline{\sigma}_T^2 = \frac{1}{T - t_0} \sum_{t=t_0+1}^{T} \big(d_t - \overline{d}_T\big)^2$$

their empirical average and variance.

Appendix B.1. Descriptive statistics

In this section we consider the no-change prediction as the benchmark; it provides the forecasts $\widehat{s}'_t = s_{t-1}$.

To get an initial feeling of whether H_0 should be rejected or not in this case, we can take a look at the quantiles of the sequence of d_t. For example, in Table D.14 of Appendix D, we see that in the sequential ridge regression with discount factors and the exponentially weighted average strategy with discount factors cases, the distribution of the differences d_t is shifted toward positive values: the 75% and 90% quantiles are larger in absolute value than the 25% and 10% ones. This is not the case for rolling or recursive OLS regressions. Additional comments on these matters are provided at the end of Section 5.1.

Appendix B.2. A general test for comparing predictive accuracies

In this section we sometimes still consider the no-change prediction as the benchmark: $\widehat{s}'_t = s_{t-1}$, but also consider other benchmarks, like rolling or recursive OLS regression, e.g., in Table 3. We recall that we aim to test the hypothesis H_0 that the difference in forecasting accuracy between the method under scrutiny and the benchmark is not significant against the alternative hypothesis H_1 that the method – the one outputting the forecasts $\widehat{s}_t$ – is significantly better, on average.

The Diebold-Mariano test with relevant assumptions. Diebold and Mariano (1995) have presented a test relying directly on the differences d_t in the forecasting errors, even though these may be serially correlated. Only mild and direct assumptions on the statistical behavior of these differences d_t need to made. Here, we state their results with a rectangular lag, as they advocate.

More precisely, they showed that under assumptions of covariance stationarity and short memory of the differences d_t, for a properly chosen truncation lag denoted by $H \geqslant 0$, the test statistics

$$S_{\mathrm{DM},H} = \sqrt{T - t_0}\, \frac{\overline{d}_T}{\sqrt{\overline{\sigma}_T^{H,2}}}\,,$$

where

$$\overline{\sigma}_T^{H,2} = \frac{1}{T - t_0} \sum_{t=t_0+1}^{T} \bigl(d_t - \overline{d}_T\bigr)^2 + \frac{2}{T - t_0} \sum_{\tau=1}^{H} \sum_{t=\tau+t_0+1}^{T} \bigl(d_t - \overline{d}_T\bigr)\bigl(d_{t-\tau} - \overline{d}_T\bigr)\,,$$

converge to a $\mathcal{N}(0,1)$ distribution under H_0 and to $+\infty$ under H_1. Diebold (2012) emphasized the generality of the method to compare the predictive accuracy of forecasts between any two methods, as long as the mild assumption stated above, namely, covariance stationarity of the differences in accuracy d_t, holds. He explains in Section 2.2 of the mentioned reference why this assumption is natural and often met in practice.

The choice of the truncation lag H was partially left open; Diebold and Mariano suggested choosing it as a function of the length of the short memory (of the autocovariance degree). Estimating by $\widehat{H}$ a proper H based on our data, then substituting its value, would lead to considering the test statistic $S_{\mathrm{DM},\widehat{H}}$, which would no longer be guaranteed to converge to a $\mathcal{N}(0,1)$ distribution under H_0. We instead take a more robust approach to reject H_0, that is, we build a more conservative test than the original test based on a good a priori value of H. Namely, we consider

$$S_{\mathrm{DM}} = \sqrt{T - t_0}\, \frac{\overline{d}_T}{\sqrt{\displaystyle\max_{H \in \{0,1,\ldots,20\}} \overline{\sigma}_T^{H,2}}}\,, \tag{B.2}$$

which is smaller than any of the corresponding original statistics $S_{\mathrm{DM},H}$. The limiting distribution under H_0 has a cumulative distribution function that is uniformly smaller than that of the $\mathcal{N}(0,1)$ distribution on positive numbers (while convergence to $+\infty$ is still achieved under H_1, at a slower rate). Note that for our data set a maximal value of 20 for H was set because it corresponds roughly to the maximal value of $\sqrt{T - t_0}$ on our data. In the following, the test described above will be referred to as the DM test.

We do not comment on the direct computation of the p-values for this DM test using quantiles of the normal distribution (which would be a conservative approach, i.e., the size and power of the test would be smaller than usual), as we rather resort to a bootstrapped approach to compute them.

Bootstrapped version. We follow here, as others have, the method and choices made by Mark and Sul (2001), Rogoff and Stavrakeva (2008), and Ince (2014). They all point out that the DM test is undersized and has less power than it should have when the conditions for its applications are not met, in particular, in the presence of forecast biases. They recommend computing its p-values based on a bootstrap procedure. Briefly, $M =$ 1 000 bootstrapped simulations of the exchange rate and fundamentals time-series are computed, the relevant forecasting procedure (e.g., the exponentially weighted average strategy with discount factors) and its benchmark (e.g., the no-change predictor or rolling OLS regression) are applied to each simulation, and the resulting DM test statistic is computed (on each). This gives rise to an (empirical) distribution for the DM test statistic under H_0. Then, the DM test statistic is computed on the original data and the bootstrapped DM p-value is obtained, in a one-sided way, based on this distribution (it equals the empirical frequency of simulated DM test statistics that are above the original DM test statistic). These bootstrapped DM p-values are found in various tables of this article in the columns labeled "DM p-value" (without any specific reference to their bootstrap origin).

We now describe the method to simulate the time-series at hand. There are slight variations between the methods considered by Mark and Sul (2001), Rogoff and Stavrakeva (2008) and Ince (2014); we chose to stick to Rogoff and Stavrakeva (2008) here. For each country and each fundamental j, the data generating process is assumed to be

$$\Delta s_t = \varepsilon_t,$$

$$\Delta f_{j,t} = \alpha_j + \beta_j t + \gamma_j f_{j,t-1} + \sum_{k=1}^{d} \delta_{j,k} \Delta s_{t-k} + \sum_{k=1}^{\ell} \zeta_{j,k} \Delta f_{j,t-k} + \varepsilon'_{j,t},$$

where s_t and $f_{j,t}$ are defined as in Section 2.1, and Δ is shorthand for the unit variations

$$\Delta s_t = s_t - s_{t-1} \qquad \text{and} \qquad \Delta f_{j,t} = f_{j,t} - f_{j,t-1}\,,$$

with ε_t and $\varepsilon'_{j,t}$ denoting the residual terms, which are modeled by random variables. That is, an autoregressive model on the evolution of the fundamentals is considered, with some external input given by the evolution of the exchange rate. For each country and each fundamental j, the parameters d and ℓ, as well as the coefficients $\delta_{j,k}$ and $\zeta_{j,k}$ of the autoregressive equation, are chosen separately. Given d and ℓ, the coefficients are estimated using OLS regression. The parameters d and ℓ selected minimize Akaike's information criterion (AIC) over the choices considered, respectively, integers 1-10 for d and 1-30 for ℓ (we set these intervals so that in our simulations, d and ℓ rarely reach the upper values 10 and 30). These two parameters may differ across countries and fundamentals. AIC is also used to decide whether to include no constant and no trend, a constant and no trend, or a constant and a trend. Note in passing that the residuals $\varepsilon'_{j,t}$ are also estimated by quantities denoted $\hat{\varepsilon}'_{j,t}$. Once the data generating processes have been all estimated, the exchange rates s_t and fundamentals $f_{j,t}$ are simulated recursively as follows.

First, we need two inputs for this recursive simulation: (i) seeds to initiate the autoregressive simulations, and (ii) values for the stochastic residuals $\varepsilon'_{j,t}$ and ε_t. The latter will be sampled with replacement among the $\hat{\varepsilon}'_{j,\tau}$ and the Δs_τ, where $\tau \in \{1, \ldots, T\}$. As for the seeds, they need to be in number $\max\{d, \ell\}$; we obtain them by sampling with replacement among the original vectors $(\Delta s_\tau, f_{1,\tau}, \ldots, f_{N,\tau})$, where $\tau \in \{1, \ldots, T\}$. Note that we use here the case-by-case resampling also considered in Rogoff and Stavrakeva (2008); we do not consider blocks of vectors.

Last, we note that we actually simulate the process until round $T + 100$ and discard the 100 first observations thus simulated, to obtain time-series of the same length T as the original series. We do so, following Rogoff and Stavrakeva (2008), to minimize the influence of the choice of seeds and make sure that the re-simulated times series are in a steady state.

Appendix B.3. A test for the case of nested forecasting equations?

The tests by West (1996), Clark and West (2006, 2007) are designed for comparing models, while we are interested here only in comparing forecasts, and not in assuming the existence of models – see Diebold (2012) and Rossi (2013) for related discussions. Furthermore, the tests by Clark and West have been designed and studied for only the rolling and recursive OLS regression cases, while here we are obviously interested in alternative methods. This is underlined in the original papers and pointed out again by Rogoff and Stavrakeva (2008).

Nonetheless, for the sake of completeness, we computed p-values associated with the test developed by Clark and West for nested models (even though we consider "forecasting equations" only and not models). These p-values should merely be considered as statistical indicators of predictability. We believe that the most significant tests being performed in our study are the DM tests mentioned above. The readers will note that in our tables, the test by Clark and West is typically less conservative than the DM one (as seen in other empirical and theoretical studies).

Mathematical description of the test. For this test we restrict our attention to the no-change prediction as the benchmark method: $\widehat{s}'_{t+1} = s_t$.

This no-change prediction is nested into the larger forecasting equation (1). In this case, assuming the existence of underlying models, Clark and West (2006) argued that the differences in forecasting abilities d_t were an unfair measure, and advocated instead to use adjusted differences, which in our context would be

$$a_t = d_t + \left(\widehat{s}_t - s_{t-1}\right)^2,$$

where $\widehat{s}_t$ is the prediction associated with the larger forecasting equation (1). We denote by

$$\overline{a}_T = \frac{1}{T - t_0 + 1} \sum_{t=t_0+1}^{T} a_t \tag{B.3}$$

the empirical average of the adjusted differences a_t and

$$\overline{S}_T^2 = \frac{1}{T - t_0 + 1} \sum_{t=t_0+1}^{T} \left(a_t - \overline{a}_T\right)^2 \tag{B.4}$$

their empirical variance. The test statistic is equal to

$$S_{\text{CW}} = \sqrt{T - t_0 + 1}\,\frac{\overline{a}_T}{\sqrt{\overline{S}_T^2}}\,. \tag{B.5}$$

The hypotheses tested are in terms of an underlying model (if we are ready to assume that one exists). Namely, H_0' : the true underlying model is the no-change model versus H_1' : the true underlying linear model uses at least one fundamental of (1). Here again, we see that the test concerns predictability, not forecastability. Under the assumption that residuals of the models form a martingale difference sequence, and provided that rolling and recursive OLS regressions are considered when selecting coefficients in (1), Clark and West (2006) showed that S_{CW} converges to a $\mathcal{N}(0,1)$ distribution under H_0' and to $+\infty$ under H_1'. The p-values associated with the test thus constructed are found in the tables in the columns labeled "CW p-value"; they are based on quantiles of the normal distribution. In this article, this test will be referred to as the CW test.

We recall that as indicated several times above, these p-values are merely indicators, and of predictability rather than of forecastability. We include them here for the sake of completeness.

Appendix C. Additional technical details

In this section, we provide

- explanations on how to run a single instance of EWA and SRidge in practice, by choosing their hyperparameters (Appendix C.1) well,
- some additional details on the numerical implementation of our machine-learning methods (Appendix C.2),
- a more complete description of the economic evaluation of our forecasting strategies, including a presentation of how to include constraints on the currency weights (Appendix C.3).

Appendix C.1. EWA and SRidge in practice: how to choose their hyperparameters

We recall that in our simulation study we do not report the performance of the EWA and SRidge methods for several well-chosen (hand-picked) sets of hyperparameters, as is usual in the literature. We instead re-use a more operational approach already considered in Devaine et al. (2013), based on a sequential grid search. We thus only report the results obtained for one run of the method. The aim of this section is to state all details of the grid-search approach followed.

For EWA, the single run selects at each time instance t, based on available data, the hyperparameters $\eta_t > 0$ and $\gamma_t > 0$ (rather than the fixed γ initially considered) to be used in (8). We chose $\kappa = 2$, which is common in the literature and allows us to have a theoretical bound given condition (10). More precisely, the parameters η_t and γ_t for the prediction of instance $t+1$ are respectively selected in the grids $\mathcal{E}/\sqrt{t}$ and $\mathcal{G}$, where

$$\begin{aligned} &\mathcal{E} = \bigl\{m \times 10^k, \;\; m \in \{1,2,5\} \text{ and } k \in \{-4,-3,-2,-1\}\bigr\} \cup \{1\}\,, \\ \text{and} \quad &\mathcal{G} = \{0\} \cup \bigl\{m \times 10^k, \;\; m \in \{1,2,5\} \text{ and } k \in \{0,1,2\}\bigr\} \cup \{1\,000\} \\ &\phantom{\mathcal{G}} = \bigl\{0,\, 1,\, 2,\, 5,\, 10,\, 20,\, 50,\, 100,\, 200,\, 500,\, 1000\bigr\}\,. \end{aligned}$$

These grids are (somewhat) evenly spaced on a logarithmic scale. To predict instance $t+1$, we resort to (8), with the pair of parameters (η, γ) in the grids $\mathcal{E}$ and $\mathcal{G}$, whose associated EWA strategy with sequence of learning parameters η, $\eta/\sqrt{1}$, …, $\eta/\sqrt{t-1}$ and fixed discount factor γ performed best overall on instances 1 to t.

We proceed similarly for SRidge: we take the same discount power $\kappa = 2$ as in the EWA case and calibrate the hyperparameters λ_t and γ_t to be used for month $t+1$ based on past data, using the same grid $\mathcal{G}$ for the γ parameters, and the grid for λ given by

$$\begin{aligned} \Lambda &= \{0\} \cup \bigl\{m \times 10^k, \;\; m \in \{1,2,5\} \text{ and } k \in \{1,2,3\}\bigr\} \cup \{10\,000\} \\ &= \bigl\{0,\, 1,\, 2,\, 5,\, 10,\, 20,\, 50,\, 100,\, 200,\, 500,\, 1\,000,\, 2\,000,\, 5\,000,\, 10\,000\bigr\}\,. \end{aligned}$$

Appendix C.2. Notes on computations and forecasts

Calculations were performed with the Scilab 5.5.0 software; the code is available upon request.

We recall that the training period for our methods was set at 120 months, which is standard in the literature, and that, correspondingly, the rolling OLS regressions were estimated with 120 months of past data at each instance. The computed coefficients of our forecasts vary greatly over time and it is difficult to discern any temporal patterns – also a feature known in the literature. This is not a surprise as our forecasting methods do not aim to estimate some model – they merely output efficient forecasts. The regularization and discount terms that are computed from past data are non-zero most of the time, and differ substantially between currencies, fundamentals used, and time periods. Most of the time the best discount factors are high – above 10 – which means that short-term trends are weighted most strongly.

We tried different grids from those in Appendix C.1. All grids were logarithmic, as recommended by machine learning theory. We found that a larger grid typically yielded small improvements in terms of RMSE with respect to the initial grid – but not always, as the RMSE of predictions can get worse. With a denser grid we may overfit the regularization and the discounting terms themselves. Also, smaller obtained RMSEs do not guarantee better DM tests being obtained; all depends on the errors of individual predictions. It appears that there is a large set of said parameters where the quality of predictions (measured by the RMSE) are qualitatively very similar, all corresponding to an improvement over the OLS methods.

Appendix C.3. Economic evaluation of forecasting strategies

We recall here the methodology introduced by Della Corte et al. (2009, 2011) and well-summarized in Li et al. (2015). We followed it scrupulously to produce our main economic evaluation results, namely, those reported in Table 5 in the "no c" lines. We also considered two additional approaches that deal with known problems in economic criterion evaluation – by constraining the weights, which led to the lines with "$c = \ldots$" in the main and certain other tables. These: Tables D.17 and D.18, show results under a summation-to-less-than-1 constraint for the weights. Below, we recall only the basic elements of the methodology for economic evaluation, so as to be able to describe the two modified approaches.

Brief description of the basic methodology for economic evaluation. We consider one domestic (US) bond, with risk-free return rate $r_{\mathrm{free},t}$, and K foreign bonds, indexed by $k = 1, \ldots, K$, with risk-free return rates in their local currencies (that would be used as UIRP fundamentals) but with risky returns $r_{k,t}$ in US dollars because (and only because) of the exchange-rate risk. The subscripts t index the time instances, namely, months in our case. At each month $t-1$, an investor forecasts the evolution of the exchange rates using our methods, and based on these forecasts, picks weights $w_t = (w_{1,t}, \ldots, w_{K,t})^{\mathrm{T}} \in \mathbb{R}^K$ determining how to invest their money in the foreign bonds for month t; these weights are real numbers. Their return in month t equals

$$\sum_{k=1}^{K} w_{k,t} r_{k,t} + \left(1 - \sum_{k=1}^{K} w_{k,t}\right) r_{\mathrm{free},t} = r_{\mathrm{free},t} + \sum_{k=1}^{K} w_{k,t} \big(r_{k,t} - r_{\mathrm{free},t}\big) \,.$$

Next, the returns $r_{k,t}$ are not yet known in month $t-1$, but their conditional means $\overline{r}_{k,t}$ and variance-covariance matrix Σ_t can be assumed to be (where conditioning is with respect to the $r_{k,t-1}$ values for month $t-1$). Indeed, the covariance matrix Σ_t used follows from an empirical estimation based on the past covariances, while the means $\overline{r}_{k,t}$

are given precisely by our forecasts. A target conditional volatility $\overline{\sigma}$ is then set and the weights are picked to optimize the following mean-variance problem:

$$\begin{aligned} \max_{w_t \in \mathbb{R}^K} \quad & r_{\mathrm{free},t} + \sum_{k=1}^{K} w_{k,t} \bigl(\overline{r}_{k,t} - r_{\mathrm{free},t} \bigr), \\ \text{subject to} \quad & w_t^{\mathrm{T}} \, \Sigma_t \, w_t \leqslant \overline{\sigma}^2. \end{aligned}$$

A closed-form solution exists when no further constraint is imposed on the weights.

Imposing further constraints on the weights. As indicated in the main text, the weights chosen as above often have large components – especially if the returns of some currencies are highly correlated – and are unstable over time. A good way to avoid such issues is to add a squared regularization term; given a parameter $c > 0$, we want to perform the optimization

$$\begin{aligned} \max_{w_t \in \mathbb{R}^K} \quad & r_{\mathrm{free},t} + \sum_{k=1}^{K} w_{k,t} \bigl(\overline{r}_{k,t} - r_{\mathrm{free},t} \bigr), \\ \text{subject to} \quad & w_t^{\mathrm{T}} \, \Sigma_t \, w_t \leqslant \overline{\sigma}^2 \\ \text{and} \quad & \sum_{k=1}^{K} w_{k,t}^2 \leqslant c. \end{aligned}$$

We provide no closed-form solution to this problem, and instead solve it numerically to obtain our results. As this constrained maximization problem is a particular case of what is called a quadratically constrained quadratic program (QCQP), its solution is equal to that of its Lagrangian dual problem, which is of the form: minimize a (nonlinear) function of finitely many variables λ_1, λ_2, ..., subject to positivity constraints on only the λ_i. See Boyd and Vandenberghe (2009, Section 5.2.4) for details on this equivalence. Now, the dual problem is easily solved numerically with the help of toolboxes from various software packages; in our case, we resorted to the `optim` routine of Scilab.

Further constraining the problem as

$$\begin{aligned} \max_{w_t \in \mathbb{R}^K} \quad & r_{\mathrm{free},t} + \sum_{k=1}^{K} w_{k,t} \bigl(\overline{r}_{k,t} - r_{\mathrm{free},t} \bigr), \\ \text{subject to} \quad & w_t^{\mathrm{T}} \, \Sigma_t \, w_t \leqslant \overline{\sigma}^2 \\ \text{and} \quad & \sum_{k=1}^{K} w_{k,t}^2 \leqslant c \\ \text{and} \quad & \sum_{k=1}^{K} w_{k,t} \leqslant 1, \end{aligned}$$

it remains a QCQP, so the methodology above still applies.

Details on these calculations (exact statements of the dual problems, code written to solve them, etc.) are available upon request.

Appendix D. Additional graphs and tables

This section provides complementary results as well as tables omitted from the main text.

Tables with results discussed in the text.

- Tables D.10–D.13 give supplementary information to the basic results presented in Section 5.
- Table D.14 gives the quantiles of the differences in forecasting error between the no-change prediction and the methods we look at.
- Table D.15 gives the results for a subsample spanning the whole range of observations from January 1980 until the end of our sample. Note that some results are not available for the PTE/USD pair as data on interest rates starts in 1983.
- Tables D.16–D.18 show robustness results for the economic evaluation of the prediction performance of the exponentially weighted average strategy with discount factors: with or without the Danish Krone, and with or without a summation-to-less-than-1 constraint on the currency weights.
- Tables D.19–D.24 show our results for the shorter period discussed in Section 5.5, namely, February 1999–December 2014.
- Table D.25 shows the results for the Taylor-rule based exchange rate fundamentals in their "decoupled" (heterogenous) form.

Comments on the weights seen in the exponentially weighted average strategy with discount factors for forecasts with all fundamentals. By construction, the exponentially weighted average strategy with discount factors has a natural interpretation as being able to identify "scapegoat" fundamentals that could have been considered by traders in some periods as in the model of Bacchetta and van Wincoop (2004). This is because based on past data and discounting, the fundamentals that are best at predicting exchange rates in a given period are given more weight while forming predictions. It is insightful therefore to investigate the behavior of these weights with time. For the sake of brevity, we show the results in Figures D.1–D.3 only for the case of predictions with *all* fundamentals for JPY/USD, USD/GBP and CHF/USD. Note that these were the setups which did not necessarily perform well in comparison to parsimonious models such as those based solely on PPP or UIRP; however, they did much better overall than the OLS methods as measured by Theil ratios and in direct comparisons.

The codes for the various fundamentals are the following: PPP refers to inflation rates, UIRP to money-market interest rates; MONEY to money stocks, and Y to industrial production, while INTCH denotes changes in money market interest rates. Lastly, "no-change" refers to the no-change forecast that also is assigned a weight in the exponentially weighted average strategy with discount factors.

The conclusion from these figures is striking: typically only one fundamental is given a predominant weight in a given period, and most of the time for very short time intervals. In most cases, the fundamentals that are given the most weight pertain to money market interest rates and money supply, and for one country only. Rarely do we observe this behavior for inflation or industrial production. It would be consistent with the scapegoat model as well as the fickle nature of expectations about the fundamentals driving the exchange rate. This is, however, open to yet another interpretation. It also may point out that different shocks to different fundamentals (news) could be driving exchange rates at different times, and – given their prevalence in our data – monetary policy and financial market shocks are the most important.

Table D.10: Relative forecasting performance of machine learning methods vs. rolling and recursive regressions based on coupled fundamentals.

Currency pair	SRidge vs.						EWA vs.					
	Rolling regression			Recursive regression			Rolling regression			Recursive regression		
	Ratio of RMSEs	DM statistic	DM p-value	Ratio of RMSEs	DM statistic	DM p-value	Ratio of RMSEs	DM statistic	DM p-value	Ratio of RMSEs	DM statistic	DM p-value
PPP fundamental												
USD/GBP	**0.9956**	1.3165	0.201	**0.9971**	0.8829	0.240	**0.9955**	1.2640	0.332	**0.9970**	1.4420	0.104
JPY/USD	**0.9942**	0.9537	0.349	**0.9990**	1.8063	0.024 **	**0.9916**	1.4361	0.236	**0.9963**	1.3909	0.128
CHF/USD	**0.9973**	1.0247	0.281	**0.9991**	0.9050	0.187	**0.9969**	1.1618	0.330	**0.9986**	1.3391	0.127
CAD/USD	1.0036	-0.5876	0.964	1.0006	-0.4685	0.830	1.0065	-0.9187	0.995	1.0035	-1.5869	0.998
SEK/USD	1.0002	-0.0373	0.859	**0.9984**	1.0210	0.204	**0.9985**	0.2664	0.825	**0.9967**	1.4879	0.097 *
DNK/USD	**0.9993**	0.1467	0.789	1.0003	-0.1096	0.702	**0.9988**	0.2587	0.834	**0.9998**	0.0826	0.768
USD/AUD	1.0031	-0.3065	0.904	**0.9974**	1.0160	0.206	1.0034	-0.3100	0.959	**0.9977**	0.8274	0.349
FRF/USD	**0.9933**	1.3725	0.103	**0.9980**	0.8185	0.223	**0.9910**	1.1881	0.241	**0.9956**	0.9072	0.338
DEM/USD	**0.9938**	1.2303	0.150	**0.9991**	0.7742	0.270	**0.9906**	1.6264	0.080 *	**0.9960**	1.5920	0.064 *
ITL/USD	**0.9985**	0.4249	0.506	**0.9993**	0.3436	0.477	**0.9933**	0.8895	0.282	**0.9941**	0.9717	0.216
NLG/USD	**0.9944**	2.1562	0.011 **	**0.9983**	1.4600	0.063 *	**0.9920**	2.0211	0.026 **	**0.9959**	1.3929	0.105
PTE/USD	**0.9963**	1.0166	0.164	**0.9994**	0.7054	0.174	**0.9860**	1.7989	0.032 **	**0.9890**	1.6708	0.037 **
UIRP fundamental												
USD/GBP	**0.9914**	0.4403	0.631	1.0033	-0.2150	0.726	**0.9847**	0.8553	0.465	**0.9965**	0.3076	0.585
JPY/USD	**0.9950**	0.8402	0.422	**0.9992**	0.2996	0.444	**0.9931**	0.8592	0.462	**0.9973**	0.4948	0.436
CHF/USD	**0.9991**	0.3607	0.641	**0.9995**	0.5994	0.326	**0.9981**	0.5465	0.596	**0.9985**	0.6273	0.351
CAD/USD	**0.9974**	0.4206	0.662	**0.9995**	0.5765	0.363	**0.9993**	0.1056	0.787	1.0014	-0.6740	0.840
SEK/USD	**0.9673**	1.8231	0.048 **	**0.9871**	1.2762	0.097 *	**0.9635**	1.7678	0.090 *	**0.9832**	1.3538	0.137
DNK/USD	**0.9885**	1.4756	0.152	**0.9997**	0.2067	0.542	**0.9895**	1.2701	0.254	1.0007	-0.1702	0.768
USD/AUD	**0.9978**	0.1953	0.783	1.0008	-0.1700	0.716	**0.9903**	0.7177	0.566	**0.9933**	0.9608	0.213
FRF/USD	**0.9934**	1.0211	0.214	**0.9980**	0.3007	0.488	**0.9899**	1.1751	0.162	**0.9945**	0.6352	0.362
DEM/USD	**0.9965**	1.0815	0.196	**0.9975**	0.6603	0.285	**0.9926**	1.1911	0.187	**0.9936**	0.8948	0.251
ITL/USD	**0.9995**	0.0796	0.535	1.0059	-0.6430	0.686	**0.9810**	1.7677	0.044 **	**0.9873**	1.6676	0.049 **
NLG/USD	**0.9978**	0.2426	0.627	1.0009	-0.1021	0.675	**0.9935**	0.7807	0.380	**0.9965**	0.7081	0.344
PTE/USD	1.0045	-0.3611	0.780	1.0001	-0.0069	0.596	1.0025	-0.2315	0.753	**0.9981**	0.1578	0.550
Monetary model fundamentals												
USD/GBP	**0.9852**	0.9504	0.662	1.0028	-0.2403	0.899	**0.9837**	1.1465	0.415	1.0013	-0.1906	0.780
JPY/USD	1.0023	-0.2941	0.978	1.0023	-0.3495	0.898	**0.9923**	0.9446	0.471	**0.9923**	0.9460	0.210
CHF/USD	**0.9986**	0.2437	0.885	**0.9952**	0.7418	0.460	**0.9976**	0.2589	0.836	**0.9942**	0.6498	0.323
CAD/USD	**0.9816**	0.8233	0.664	1.0033	-0.4516	0.883	**0.9824**	0.7388	0.507	1.0042	-0.6774	0.817
SEK/USD	**0.9859**	0.9182	0.600	1.0090	-0.7431	0.956	**0.9764**	1.8596	0.120	**0.9993**	0.0869	0.648
DNK/USD	**0.9860**	1.5671	0.238	**0.9950**	1.1239	0.260	**0.9937**	0.6450	0.552	1.0027	-0.5254	0.798
USD/AUD	**0.9831**	0.9608	0.601	**0.9964**	1.1161	0.275	**0.9839**	0.7781	0.573	**0.9972**	0.3462	0.467
FRF/USD	**0.9827**	1.3614	0.157	**0.9847**	1.0018	0.247	**0.9672**	1.6177	0.073 *	**0.9692**	1.6777	0.035 **
DEM/USD	**0.9698**	2.5938	0.004 ***	**0.9835**	1.9431	0.040 **	**0.9530**	2.3338	0.010 ***	**0.9664**	1.4955	0.052 *
ITL/USD	**0.9925**	0.4679	0.545	**0.9926**	0.8150	0.280	**0.9900**	0.6796	0.536	**0.9902**	0.9250	0.295
NLG/USD	**0.9975**	0.2524	0.630	**0.9961**	0.4855	0.335	1.0053	-0.3381	0.767	1.0038	-0.2802	0.570
PTE/USD	1.0184	-0.6059	0.927	1.0081	-0.5446	0.881	1.0129	-0.7568	0.888	1.0027	-0.1971	0.660
All fundamentals												
USD/GBP	**0.9510**	2.4704	0.258	**0.9848**	1.3841	0.418	**0.9488**	2.6541	0.144	**0.9825**	1.3737	0.349
JPY/USD	**0.9852**	0.8267	0.938	**0.9968**	0.5586	0.770	**0.9765**	1.4380	0.781	**0.9880**	1.2918	0.403
CHF/USD	**0.9833**	1.9802	0.320	**0.9921**	1.1389	0.271	**0.9788**	2.0200	0.299	**0.9876**	1.3177	0.201
CAD/USD	**0.9791**	0.9818	0.878	1.0016	-0.2222	0.916	**0.9803**	0.8855	0.855	1.0029	-0.4781	0.925
SEK/USD	**0.8616**	1.3058	0.642	**0.9355**	1.0770	0.467	**0.8438**	1.2901	0.683	**0.9162**	1.0959	0.479
DNK/USD	**0.9707**	1.9689	0.452	**0.9925**	1.6196	0.291	**0.9751**	1.6270	0.615	**0.9971**	0.5758	0.774
USD/AUD	**0.9719**	1.2854	0.810	**0.9885**	1.5011	0.308	**0.9712**	1.2579	0.791	**0.9878**	1.1158	0.394
FRF/USD	**0.9563**	1.9726	0.083 *	**0.9786**	0.9804	0.288	**0.9349**	1.9597	0.066 *	**0.9567**	1.3087	0.140
DEM/USD	**0.9519**	2.6961	0.032 **	**0.9737**	2.2049	0.039 **	**0.9347**	2.4342	0.065 *	**0.9562**	1.8537	0.092 *
ITL/USD	**0.9767**	0.9464	0.608	**0.9691**	1.2010	0.288	**0.9735**	1.0142	0.626	**0.9659**	1.1458	0.369
NLG/USD	**0.9816**	1.0200	0.620	**0.9914**	0.8213	0.496	**0.9854**	0.7195	0.705	**0.9953**	0.3713	0.651
PTE/USD	**0.9856**	0.5718	0.611	**0.9586**	1.4498	0.148	**0.9823**	0.5867	0.620	**0.9554**	1.1561	0.273

Notes: This table presents the comparison of forecasting performance between, respectively, SRidge and EWA vs. the rolling and recursive OLS for coupled fundamentals. Columns 1-6 show the comparison between SRidge and OLS methods, while columns 7-12 for EWA vs. OLS. For each comparison, the Theil ratio (RMSE of the given method / RMSE of the compared) and the DM statistic and its bootstrapped p-value are shown. The DM test is a one-sided test of equal out-of-sample prediction accuracy (H_0) against superior out-of-sample prediction accuracy (H_1) of the machine-learning method considered against the OLS methods. ***, **, and * denote statistical significance at the 1 %, 5 %, and 10 % levels. Theil ratios < 1 are indicated in bold. Sample period for the exchange rates: March 1973 – December 2014 unless shorter (indicated in Appendix A).

Table D.11: Directional predictions of the exchange rates based on coupled fundamentals.

Currency pair	Rolling regression			Recursive regression			Sridge			EWA		
	Proportion of changes predicted	DM statistic	DM p-value	Proportion of changes predicted	DM statistic	DM p-value	Proportion of changes predicted	DM statistic	DM p-value	Proportion of changes predicted	DM statistic	DM p-value
PPP fundamental												
USD/GBP	**0.501**	0.0385	0.491	0.493	-0.2215	0.565	**0.507**	0.1924	0.392	0.493	-0.2215	0.561
JPY/USD	0.467	-1.1724	0.878	0.478	-0.7410	0.757	**0.554**	2.1120	0.008 ***	**0.509**	0.3016	0.267
CHF/USD	**0.504**	0.1330	0.441	0.486	-0.4875	0.697	0.486	-0.4875	0.667	0.486	-0.4875	0.686
CAD/USD	**0.517**	0.6664	0.215	0.491	-0.3508	0.627	0.491	-0.3508	0.637	0.478	-0.8586	0.861
SEK/USD	**0.514**	0.3831	0.313	0.486	-0.3768	0.656	**0.507**	0.2160	0.401	0.441	-1.8029	0.981
DNK/USD	**0.504**	0.1227	0.485	**0.507**	0.1916	0.424	**0.564**	2.2004	0.005 ***	0.491	-0.2737	0.553
USD/AUD	**0.522**	0.5909	0.228	0.447	-1.6911	0.956	0.447	-1.6911	0.954	0.447	-1.5597	0.944
FRF/USD	0.466	-0.5013	0.708	**0.545**	0.6949	0.218	**0.513**	0.2284	0.377	**0.545**	0.6949	0.250
DEM/USD	0.487	-0.3266	0.692	**0.529**	0.6309	0.345	**0.529**	0.6309	0.316	**0.540**	0.8492	0.337
ITL/USD	**0.534**	0.5861	0.283	0.455	-0.8170	0.863	0.450	-0.9118	0.874	0.455	-0.8170	0.872
NLG/USD	0.471	-0.5985	0.783	0.471	-0.5328	0.783	0.471	-0.5328	0.771	0.492	-0.1520	0.751
PTE/USD	**0.524**	0.4381	0.492	0.481	-0.3120	0.812	0.481	-0.3120	0.805	0.481	-0.3120	0.809
UIRP fundamental												
USD/GBP	**0.514**	0.5527	0.259	**0.533**	1.2562	0.089 *	**0.504**	0.1455	0.423	0.465	-1.3720	0.919
JPY/USD	0.491	-0.2988	0.600	**0.520**	0.6265	0.219	**0.517**	0.5416	0.250	0.478	-0.7171	0.704
CHF/USD	**0.512**	0.4190	0.323	0.482	-0.5189	0.712	0.482	-0.5189	0.687	**0.512**	0.3727	0.249
CAD/USD	**0.535**	1.1080	0.124	**0.507**	0.1946	0.426	**0.507**	0.1946	0.418	0.483	-0.4852	0.553
SEK/USD	**0.500**	0.0000	0.422	0.489	-0.3343	0.579	**0.542**	1.5619	0.044 **	0.466	-1.0618	0.787
DNK/USD	**0.538**	1.2535	0.094 *	**0.530**	0.9022	0.164	**0.530**	0.9022	0.163	0.467	-0.9952	0.783
USD/AUD	**0.508**	0.3073	0.328	0.445	-2.1153	0.984	**0.533**	1.2218	0.079 *	0.436	-2.4375	0.981
FRF/USD	**0.550**	1.0082	0.141	**0.571**	1.4934	0.056 *	**0.571**	1.7470	0.028 **	0.423	-1.6341	0.924
DEM/USD	0.471	-0.6314	0.760	0.466	-0.7603	0.801	0.466	-0.7603	0.801	**0.524**	0.5189	0.389
ITL/USD	0.492	-0.1391	0.550	0.487	-0.2371	0.604	0.481	-0.3022	0.602	0.487	-0.2371	0.576
NLG/USD	0.455	-0.8298	0.838	0.476	-0.4540	0.689	**0.561**	1.5501	0.054 *	**0.524**	0.4540	0.446
PTE/USD	0.465	-0.4644	0.695	0.479	-0.2975	0.617	0.479	-0.2975	0.611	0.479	-0.2975	0.659
Monetary model fundamentals												
USD/GBP	**0.514**	0.4750	0.285	**0.541**	1.2704	0.073 *	**0.535**	1.1756	0.086 *	**0.525**	0.9346	0.080 *
JPY/USD	0.476	-0.7489	0.798	0.479	-0.6835	0.795	**0.516**	0.5121	0.305	**0.568**	2.4519	0.007 ***
CHF/USD	**0.516**	0.5688	0.250	**0.536**	1.2759	0.067 *	**0.538**	1.3542	0.056 *	**0.530**	0.9899	0.136
CAD/USD	0.482	-0.6747	0.772	**0.500**	0.0000	0.453	**0.503**	0.0925	0.458	0.482	-0.6720	0.733
SEK/USD	0.476	-0.8706	0.810	0.482	-0.5908	0.707	0.487	-0.4972	0.690	**0.518**	0.6926	0.282
DNK/USD	0.469	-0.7879	0.814	0.471	-0.7988	0.822	0.497	-0.0917	0.512	0.466	-1.0449	0.886
USD/AUD	**0.511**	0.4179	0.295	**0.511**	0.4168	0.270	**0.514**	0.4535	0.297	**0.540**	1.3903	0.048 **
FRF/USD	0.467	-0.6763	0.797	0.475	-0.5034	0.752	**0.517**	0.3654	0.359	**0.525**	0.5484	0.249
DEM/USD	0.435	-1.3965	0.937	0.452	-0.9529	0.858	**0.548**	1.2480	0.113	**0.548**	0.8565	0.091 *
ITL/USD	0.474	-0.4744	0.690	0.481	-0.4249	0.649	**0.513**	0.2732	0.390	**0.538**	0.8854	0.124
NLG/USD	0.492	-0.1438	0.688	**0.513**	0.2017	0.568	**0.577**	1.6343	0.094 *	0.466	-0.8248	0.632
PTE/USD	**0.529**	0.3462	0.354	0.486	-0.1644	0.521	0.471	-0.4729	0.696	0.429	-1.0750	0.787
All fundamentals												
USD/GBP	0.496	-0.1469	0.516	0.493	-0.2330	0.546	**0.535**	1.0098	0.135	**0.522**	0.6154	0.180
JPY/USD	**0.537**	1.2802	0.083 *	0.495	-0.1728	0.564	**0.534**	1.2103	0.093 *	**0.571**	2.7985	0.002 ***
CHF/USD	0.481	-0.6283	0.847	**0.536**	1.2186	0.244	**0.538**	1.3417	0.068 *	**0.541**	1.4025	0.056 *
CAD/USD	**0.534**	1.3369	0.113	**0.500**	0.0000	0.643	0.484	-0.5375	0.701	0.450	-1.7858	0.931
SEK/USD	0.487	-0.4661	0.651	**0.503**	0.0776	0.453	**0.503**	0.0778	0.429	**0.505**	0.1947	0.460
DNK/USD	**0.506**	0.1976	0.406	**0.500**	0.0000	0.464	0.489	-0.3858	0.684	0.446	-1.6878	0.943
USD/AUD	**0.514**	0.4478	0.333	**0.500**	0.0000	0.496	0.480	-0.6106	0.714	**0.520**	0.7277	0.110
FRF/USD	**0.500**	0.0000	0.530	**0.517**	0.3507	0.439	**0.542**	0.8541	0.233	**0.583**	1.8516	0.025 **
DEM/USD	0.429	-1.4965	0.961	0.446	-0.9310	0.895	**0.514**	0.3155	0.480	**0.610**	2.4179	0.015 **
ITL/USD	**0.538**	0.8281	0.166	**0.500**	0.0000	0.448	0.423	-1.4450	0.949	0.487	-0.2305	0.558
NLG/USD	**0.519**	0.3282	0.453	**0.529**	0.6907	0.342	**0.593**	2.1752	0.019 **	**0.524**	0.6554	0.258
PTE/USD	0.443	-0.7467	0.809	0.357	-2.1654	0.984	0.443	-0.9625	0.827	**0.500**	0.0000	0.487

Notes: This table presents the results of forecasting the direction of change of the end-of-month exchange rate 1-month ahead by the rolling OLS, recursive OLS, SRidge and EWA methods for based on coupled fundamentals. For each method, the share of correctly forecasted changes, the DM statistic and its bootstrapped p-value are shown. The DM tests is a one-sided test of equal out-of-sample prediction accuracy (H_0) against superior out-of-sample prediction accuracy (H_1) for the methods considered against the benchmark of a 50 % success rate. ***, **, and * denote statistical significance at the 1 %, 5 %, and 10 % levels. Shares > 0.5 are indicated in bold. Sample period for the exchange rates: March 1973 – December 2014 unless shorter (indicated in Appendix A).

Table D.12: Relative forecasting performance on the real-time data set of machine-learning methods vs. rolling and recursive regressions based on coupled fundamentals.

Currency pair	SRidge vs.						EWA vs.					
	Rolling regression			Recursive regression			Rolling regression			Recursive regression		
	Ratio of RMSEs	DM statistic	DM p-value	Ratio of RMSEs	DM statistic	DM p-value	Ratio of RMSEs	DM statistic	DM p-value	Ratio of RMSEs	DM statistic	DM p-value
PPP fundamental												
USD/GBP	**0.9871**	0.5723	0.781	**0.9957**	0.1864	0.701	**0.9960**	0.2503	0.917	1.0047	-0.3038	0.918
JPY/USD	**0.9749**	1.6693	0.199	**0.9834**	1.9591	0.022 **	**0.9762**	1.6026	0.265	**0.9846**	2.0033	0.020 **
CHF/USD	**0.9727**	1.2840	0.390	**0.9922**	0.6511	0.531	**0.9635**	1.8364	0.188	**0.9829**	1.8253	0.083 *
CAD/USD	**0.9956**	0.2401	0.882	1.0024	-0.1289	0.863	**0.9933**	0.3510	0.876	1.0001	-0.0061	0.868
SEK/USD	**0.9756**	0.7510	0.675	**0.9873**	0.3774	0.579	**0.9734**	1.2039	0.444	**0.9851**	0.6260	0.548
DNK/USD	**0.9767**	0.8632	0.695	**0.9984**	0.0894	0.820	**0.9704**	0.9292	0.713	**0.9920**	0.6698	0.673
USD/AUD	1.0007	-0.0419	0.927	1.0153	-0.7211	0.938	**0.9930**	0.3423	0.843	1.0075	-0.4080	0.901
UIRP fundamental												
USD/GBP	**0.9671**	1.2022	0.517	**0.9834**	1.9048	0.042 **	**0.9666**	1.2175	0.567	**0.9828**	1.8546	0.058 *
JPY/USD	**0.9244**	1.5055	0.367	**0.9814**	1.1048	0.291	**0.9203**	1.6042	0.385	**0.9769**	1.2853	0.302
CHF/USD	**0.9732**	1.4927	0.327	**0.9915**	0.7455	0.469	**0.9721**	1.4891	0.391	**0.9904**	0.7868	0.564
CAD/USD	**0.9785**	1.5810	0.351	**0.9971**	0.4522	0.700	**0.9741**	1.8677	0.243	**0.9926**	1.2498	0.329
SEK/USD	**0.9680**	1.9863	0.175	**0.9763**	1.1811	0.284	**0.9659**	1.8016	0.301	**0.9742**	1.2557	0.348
DNK/USD	**0.9682**	1.4452	0.440	**0.9753**	1.2400	0.235	**0.9607**	1.4333	0.514	**0.9677**	1.2469	0.342
USD/AUD	**0.9569**	2.1860	0.107	**0.9775**	1.1990	0.300	**0.9504**	2.4313	0.054 *	**0.9708**	1.2877	0.291
Monetary model fundamentals												
USD/GBP	**0.8899**	1.2895	0.824	**0.9600**	1.3346	0.408	**0.8954**	1.2876	0.776	**0.9659**	1.4362	0.292
JPY/USD	**0.9254**	1.7243	0.524	**0.9971**	0.1470	0.905	**0.9160**	1.9044	0.376	**0.9870**	0.6587	0.681
CHF/USD	**0.9409**	1.6310	0.636	**0.9890**	0.9857	0.577	**0.9477**	1.5354	0.650	**0.9961**	0.3536	0.832
CAD/USD	**0.9286**	1.9998	0.399	**0.9908**	0.4283	0.845	**0.9176**	1.9259	0.437	**0.9790**	1.5416	0.229
SEK/USD	**0.9082**	2.1105	0.357	**0.9848**	0.5040	0.805	**0.8851**	1.7055	0.510	**0.9597**	1.1853	0.378
DNK/USD	**0.9391**	1.8683	0.274	**0.9897**	0.6940	0.563	**0.9427**	1.5686	0.643	**0.9935**	0.2611	0.929
USD/AUD	**0.9171**	2.1485	0.273	**0.9811**	0.8950	0.521	**0.9055**	2.0108	0.300	**0.9687**	1.2467	0.278
All fundamentals												
USD/GBP	**0.8705**	2.2938	0.808	**0.9531**	1.2933	0.780	**0.8752**	2.1894	0.830	**0.9582**	1.0960	0.843
JPY/USD	**0.8028**	2.7486	0.629	**0.9339**	1.5578	0.586	**0.7974**	2.8075	0.580	**0.9277**	1.6331	0.503
CHF/USD	**0.7488**	2.1951	0.754	**0.8907**	2.6971	0.080 *	**0.7522**	2.1747	0.754	**0.8947**	2.6420	0.100 *
CAD/USD	**0.8807**	3.1080	0.376	**0.9635**	1.4850	0.638	**0.8701**	3.0716	0.398	**0.9519**	1.9354	0.347
SEK/USD	**0.8678**	1.9290	0.904	**0.9247**	1.8048	0.465	**0.8493**	1.8351	0.926	**0.9050**	2.2199	0.187
DNK/USD	**0.8469**	2.8740	0.479	**0.9049**	2.9110	0.042 **	**0.8442**	3.2012	0.384	**0.9020**	2.1731	0.398
USD/AUD	**0.8243**	2.3792	0.766	**0.9428**	2.0004	0.342	**0.8153**	2.3782	0.757	**0.9325**	2.4130	0.129

Notes: This table presents the comparison of forecasting performance between, respectively, SRidge and EWA vs. the rolling and recursive OLS for the real-time data set, based on decoupled fundamentals. Columns 1-6 show the comparison between SRidge and OLS methods, while columns 7-12 for EWA vs. OLS. For each comparison, the Theil ratio (RMSE of the given method / RMSE of the compared) and the DM statistic and its bootstrapped p-value are shown. The DM test is a one-sided test of equal out-of-sample prediction accuracy (H_0) against superior out-of-sample prediction accuracy (H_1) of the machine-learning method considered against the OLS methods. ***, **, and * denote statistical significance at the 1 %, 5 %, and 10 % levels. Theil ratios < 1 are indicated in bold. Sample period for the exchange rates: February 1999 – December 2014.

Table D.13: Directional predictions of the exchange rates for the real-time data set based on decoupled fundamentals.

Currency pair	Rolling regression			Recursive regression			Sridge			EWA		
	Proportion of changes predicted	DM statistic	DM p-value	Proportion of changes predicted	DM statistic	DM p-value	Proportion of changes predicted	DM statistic	DM p-value	Proportion of changes predicted	DM statistic	DM p-value
PPP fundamental												
USD/GBP	**0.544**	1.1182	0.107	**0.563**	1.6040	0.041 **	**0.532**	0.6838	0.236	0.424	-1.8539	0.974
JPY/USD	0.475	-0.5114	0.732	0.481	-0.4611	0.714	**0.519**	0.4285	0.322	**0.544**	0.9954	0.197
CHF/USD	**0.513**	0.2631	0.368	0.487	-0.2189	0.567	0.468	-0.7292	0.761	0.468	-0.5603	0.619
CAD/USD	**0.525**	0.5614	0.250	0.468	-0.7199	0.791	0.494	-0.1562	0.578	**0.525**	0.6373	0.273
SEK/USD	**0.563**	1.5600	0.054 *	**0.544**	0.8997	0.214	**0.532**	0.7779	0.207	0.475	-0.5524	0.605
DNK/USD	**0.529**	0.7241	0.168	**0.516**	0.3275	0.318	0.490	-0.1837	0.537	0.465	-0.7412	0.797
USD/AUD	0.481	-0.4183	0.708	**0.519**	0.3338	0.456	**0.500**	0.0000	0.589	0.392	-2.4317	0.987
UIRP fundamental												
USD/GBP	0.494	-0.1591	0.590	0.494	-0.1591	0.585	**0.519**	0.4634	0.292	**0.538**	0.9026	0.161
JPY/USD	**0.506**	0.0939	0.473	0.494	-0.1265	0.544	**0.551**	0.9673	0.157	**0.513**	0.2598	0.475
CHF/USD	**0.519**	0.4424	0.258	**0.506**	0.1444	0.392	0.468	-0.6746	0.719	0.475	-0.4356	0.626
CAD/USD	0.449	-1.2795	0.906	0.481	-0.3808	0.657	**0.500**	0.0000	0.488	**0.525**	0.5991	0.270
SEK/USD	**0.513**	0.2525	0.381	0.443	-1.3794	0.928	**0.506**	0.1303	0.443	**0.513**	0.2802	0.327
DNK/USD	0.490	-0.1837	0.571	0.477	-0.5038	0.718	0.497	-0.0647	0.483	**0.542**	0.8880	0.128
USD/AUD	0.399	-1.8345	0.974	0.487	-0.3183	0.615	0.462	-0.8305	0.803	0.475	-0.6006	0.715
Monetary model fundamentals												
USD/GBP	0.468	-0.7179	0.769	0.437	-1.1840	0.911	**0.500**	0.0000	0.482	0.462	-0.7760	0.819
JPY/USD	**0.544**	1.1182	0.112	**0.614**	2.9181	0.000 ***	**0.557**	1.4414	0.043 **	**0.551**	1.1279	0.068 *
CHF/USD	**0.513**	0.2750	0.319	0.487	-0.1827	0.502	0.449	-1.0300	0.808	0.424	-1.9190	0.951
CAD/USD	0.475	-0.5902	0.770	0.418	-2.0970	0.993	0.449	-1.2166	0.918	0.462	-0.7159	0.843
SEK/USD	0.481	-0.3709	0.662	0.462	-0.6959	0.801	**0.500**	0.0000	0.498	**0.538**	0.9254	0.113
DNK/USD	**0.542**	0.9022	0.153	**0.510**	0.2161	0.407	**0.510**	0.1507	0.371	**0.523**	0.4224	0.325
USD/AUD	**0.506**	0.1575	0.481	0.494	-0.1393	0.644	0.462	-0.9427	0.849	0.449	-1.1824	0.838
All fundamentals												
USD/GBP	**0.519**	0.4777	0.291	**0.570**	1.1277	0.113	**0.500**	0.0000	0.474	0.487	-0.2790	0.607
JPY/USD	**0.506**	0.1349	0.533	**0.538**	0.8012	0.278	**0.551**	1.2419	0.096 *	**0.557**	1.4067	0.056 *
CHF/USD	**0.551**	1.0902	0.101	**0.506**	0.1149	0.436	0.481	-0.3715	0.606	0.418	-2.0399	0.976
CAD/USD	0.475	-0.4994	0.816	0.487	-0.3058	0.792	0.443	-1.4258	0.942	0.437	-1.3736	0.946
SEK/USD	0.449	-1.1006	0.898	0.430	-1.7674	0.981	**0.519**	0.3970	0.329	**0.544**	1.1122	0.072 *
DNK/USD	0.490	-0.2067	0.583	0.439	-1.0002	0.856	**0.510**	0.1493	0.396	**0.523**	0.3804	0.399
USD/AUD	0.487	-0.3183	0.770	**0.538**	0.6922	0.370	0.462	-0.9427	0.875	0.456	-1.1182	0.850

Notes: This table presents the results of forecasting the direction of change of the end-of-month exchange rate 1-month ahead by the rolling OLS, recursive OLS, SRidge and EWA methods for the real-time data set, based on decoupled fundamentals. For each method, the share of correctly forecasted changes, the DM statistic and its bootstrapped p-value are shown. The DM tests is a one-sided test of equal out-of-sample prediction accuracy (H_0) against superior out-of-sample prediction accuracy (H_1) of the methods considered against the benchmark of a 50 % success rate. ***, **, and * denote statistical significance at the 1 %, 5 %, and 10 % levels. Shares > 0.5 are indicated in bold.

Table D.14: Distributions of the differences in forecasting error between the no-change prediction and the method under scrutiny based on decoupled fundamentals.

Currency pair	Rolling regression					Recursive regression					SRidge					EWA				
	10%	25%	50%	75%	90%	10%	25%	50%	75%	90%	10%	25%	50%	75%	90%	10%	25%	50%	75%	90%
PPP fundamental																				
USD/GBP	-1.32	-0.45	-0.01	0.41	1.00	-0.55	-0.14	0.00	0.12	0.50	-0.21	-0.06	0.00	0.04	0.25	-1.47	-0.38	0.00	0.37	1.65
JPY/USD	-2.32	-0.78	-0.06	0.53	1.75	-0.30	-0.13	-0.01	0.10	0.26	-0.02	0.00	0.00	0.01	0.02	-0.90	-0.11	0.00	0.24	1.06
CHF/USD	-1.81	-0.51	0.00	0.43	1.48	-0.78	-0.25	0.00	0.26	0.85	-0.14	-0.03	0.00	0.06	0.21	-1.34	-0.29	0.00	0.42	1.76
CAD/USD	-0.65	-0.21	0.00	0.19	0.76	-0.26	-0.11	-0.01	0.10	0.27	-0.06	-0.02	0.00	0.02	0.05	-0.05	-0.01	0.00	0.01	0.04
SEK/USD	-2.84	-1.02	-0.07	0.90	2.78	-1.11	-0.49	-0.03	0.42	1.28	-0.33	-0.09	0.00	0.08	0.37	-1.19	-0.23	0.00	0.32	1.41
DNK/USD	-2.28	-0.82	-0.02	0.63	1.56	-1.27	-0.60	0.01	0.45	1.24	-0.08	-0.01	0.00	0.03	0.10	-1.19	-0.30	0.03	0.46	1.57
USD/AUD	-2.25	-1.13	-0.03	0.92	2.66	-0.92	-0.36	-0.01	0.28	0.71	-0.24	-0.08	0.00	0.04	0.16	-1.00	-0.05	0.00	0.03	0.48
FRF/USD	-2.39	-0.81	-0.04	0.63	2.06	-1.17	-0.44	0.01	0.31	1.11	-0.26	-0.04	0.00	0.04	0.30	-1.63	-0.55	0.10	0.68	1.48
DEM/USD	-2.05	-0.71	-0.04	0.41	1.50	-0.44	-0.18	0.00	0.09	0.27	-0.11	-0.02	0.01	0.06	0.16	-1.53	-0.33	0.09	0.98	1.76
ITL/USD	-2.38	-0.83	-0.04	0.47	2.28	-2.17	-0.76	-0.16	0.66	2.39	-1.67	-0.57	-0.11	0.60	1.86	-1.46	-0.35	0.03	0.83	3.15
NLG/USD	-2.24	-0.53	-0.01	0.32	1.30	-0.29	-0.11	-0.01	0.06	0.15	-0.11	-0.02	0.00	0.05	0.14	-1.46	-0.39	0.05	0.90	1.85
PTE/USD	-5.02	-2.09	-0.21	1.54	4.88	-4.43	-1.38	-0.09	0.99	4.45	-3.72	-1.24	-0.13	1.12	4.33	-2.72	-0.70	0.00	0.75	3.94
UIRP fundamental																				
USD/GBP	-2.58	-0.64	-0.03	0.54	2.59	-0.92	-0.25	0.00	0.26	1.36	-0.29	-0.06	0.00	0.05	0.42	-2.26	-0.49	0.00	0.35	2.46
JPY/USD	-2.52	-0.92	0.03	0.97	3.19	-1.14	-0.27	0.00	0.39	1.64	-0.23	-0.02	0.00	0.04	0.25	-1.57	-0.08	0.00	0.16	1.78
CHF/USD	-3.41	-0.68	0.02	0.79	2.76	-1.76	-0.32	0.01	0.65	2.44	-0.41	-0.06	0.00	0.12	0.54	-1.67	-0.02	0.00	0.09	2.45
CAD/USD	-0.83	-0.26	0.02	0.29	0.69	-0.28	-0.10	0.00	0.10	0.24	-0.02	0.00	0.00	0.00	0.01	-0.05	-0.01	0.00	0.00	0.02
SEK/USD	-3.89	-1.40	-0.05	0.76	2.76	-2.09	-0.55	-0.03	0.21	1.55	-0.68	-0.12	0.00	0.10	0.72	-2.21	-0.57	0.00	0.38	2.06
DNK/USD	-2.66	-0.68	-0.02	0.47	1.98	-1.49	-0.39	-0.01	0.39	1.37	-0.05	-0.01	0.00	0.01	0.08	-1.92	-0.28	0.00	0.62	2.33
USD/AUD	-3.84	-1.13	-0.03	1.04	3.13	-1.60	-0.58	0.00	0.49	1.23	-0.63	-0.19	0.00	0.12	0.36	-0.26	0.00	0.00	0.00	0.08
FRF/USD	-3.36	-1.38	-0.01	0.62	2.26	-2.64	-1.02	-0.09	0.93	2.37	-0.92	-0.15	0.02	0.25	0.94	-3.57	-1.19	0.21	1.29	3.21
DEM/USD	-2.15	-0.74	-0.04	0.54	1.82	-1.10	-0.23	0.02	0.45	1.26	-0.46	-0.10	0.03	0.25	0.69	-3.47	-1.06	0.12	1.72	3.38
ITL/USD	-4.11	-1.69	-0.16	1.06	3.89	-3.51	-1.28	-0.20	0.88	2.82	-2.75	-0.82	-0.05	0.53	2.35	-3.90	-1.20	0.00	1.69	5.56
NLG/USD	-4.07	-1.02	0.00	1.04	3.35	-0.83	-0.21	0.04	0.39	1.07	-0.28	-0.05	0.01	0.15	0.31	-3.18	-1.01	0.07	1.61	3.64
PTE/USD	-2.10	-0.92	-0.11	0.67	2.60	-1.45	-0.78	-0.08	0.45	1.35	-0.70	-0.30	-0.02	0.26	0.74	-0.80	-0.28	0.00	0.22	0.57
Monetary model fundamentals																				
USD/GBP	-4.29	-1.28	-0.09	0.66	2.76	-2.29	-0.71	0.01	0.45	2.22	-1.32	-0.27	0.00	0.25	1.28	-2.49	-0.87	0.00	0.81	2.58
JPY/USD	-4.37	-1.04	-0.01	1.12	4.69	-3.85	-1.12	-0.01	1.00	4.25	-2.31	-0.54	0.01	0.67	2.47	-3.23	-0.90	0.00	1.18	3.61
CHF/USD	-5.61	-1.74	-0.09	1.44	4.82	-4.71	-1.25	0.01	1.43	4.21	-1.33	-0.37	0.01	0.50	1.36	-1.76	-0.48	0.01	0.58	1.45
CAD/USD	-1.86	-0.64	-0.01	0.36	1.09	-0.71	-0.26	-0.02	0.18	0.59	-0.11	-0.01	0.00	0.01	0.09	-0.09	-0.03	0.00	0.02	0.06
SEK/USD	-6.77	-2.37	-0.35	0.65	2.93	-3.05	-0.83	-0.05	0.42	1.35	-0.85	-0.11	0.00	0.17	0.72	-2.41	-0.79	0.00	0.71	2.45
DNK/USD	-5.15	-1.51	-0.14	0.52	1.76	-2.26	-0.66	-0.01	0.37	0.99	-0.02	-0.01	0.00	0.01	0.01	-0.80	-0.28	-0.01	0.21	0.67
USD/AUD	-4.72	-1.57	-0.01	1.03	3.66	-2.00	-0.67	-0.02	0.45	1.50	-0.24	-0.02	0.00	0.02	0.08	-1.67	-0.46	-0.01	0.43	1.49
FRF/USD	-5.48	-2.57	-0.12	1.18	3.26	-5.31	-2.22	-0.17	0.73	3.58	-1.30	-0.38	0.02	0.33	0.99	-2.97	-1.16	0.01	0.75	2.71
DEM/USD	-5.79	-1.40	0.08	1.73	3.64	-4.18	-0.94	0.08	1.47	3.15	-1.63	-0.22	0.01	0.45	1.18	-3.96	-1.69	0.01	1.47	4.39
ITL/USD	-4.61	-1.98	-0.40	0.95	3.87	-3.23	-1.41	-0.16	0.83	3.42	-1.20	-0.41	0.00	0.26	0.83	-4.10	-1.15	0.00	1.09	3.50
NLG/USD	-6.38	-2.11	-0.31	0.89	3.88	-3.81	-1.03	0.00	0.82	3.10	-0.57	-0.09	0.02	0.24	0.62	-3.00	-0.96	0.00	1.05	3.33
PTE/USD	-6.51	-1.60	-0.21	1.38	4.32	-3.80	-1.54	-0.29	0.42	1.30	-1.77	-0.41	-0.06	0.14	0.87	-2.08	-0.54	-0.01	0.66	1.76
All fundamentals																				
USD/GBP	-9.18	-2.66	-0.16	1.94	5.54	-3.11	-0.84	-0.07	0.66	2.94	-0.64	-0.16	0.00	0.19	0.84	-3.09	-0.93	-0.03	0.69	3.44
JPY/USD	-7.77	-2.95	-0.08	2.01	7.04	-5.78	-2.35	-0.05	1.99	5.71	-1.58	-0.38	0.02	0.55	2.03	-3.62	-1.08	0.01	1.21	3.46
CHF/USD	-11.91	-4.06	-0.32	2.19	6.89	-7.11	-2.06	-0.04	1.68	5.74	-0.70	-0.14	0.01	0.15	0.52	-3.87	-1.23	0.01	1.39	4.18
CAD/USD	-2.65	-0.88	0.00	0.64	1.88	-0.82	-0.28	-0.01	0.22	0.65	-0.06	-0.01	0.00	0.01	0.05	-0.09	-0.03	0.00	0.01	0.05
SEK/USD	-9.47	-3.49	-0.20	1.70	5.17	-4.71	-1.63	-0.14	1.18	4.07	-0.35	-0.08	0.00	0.06	0.34	-3.39	-1.40	0.01	1.37	3.72
DNK/USD	-7.65	-2.45	-0.13	1.43	5.47	-3.15	-0.95	-0.01	0.73	2.79	-0.04	-0.01	0.00	0.01	0.02	-0.53	-0.18	-0.01	0.12	0.33
USD/AUD	-9.14	-3.09	-0.15	2.00	5.79	-4.24	-1.31	0.00	0.86	2.52	-0.46	-0.11	0.00	0.07	0.26	-2.80	-0.26	0.00	0.17	1.44
FRF/USD	-8.52	-3.10	0.06	1.48	5.26	-6.58	-2.19	0.00	1.80	4.44	-1.53	-0.27	0.00	0.18	0.92	-4.64	-1.59	0.01	1.47	4.52
DEM/USD	-7.86	-2.38	-0.02	1.80	6.15	-5.68	-1.89	-0.05	1.75	5.06	-1.01	-0.09	0.00	0.11	0.83	-4.99	-1.86	0.05	1.95	4.78
ITL/USD	-6.21	-2.47	-0.16	1.24	6.67	-5.77	-1.98	-0.16	1.25	6.08	-1.55	-0.54	-0.06	0.33	1.33	-5.39	-1.67	0.03	1.38	6.05
NLG/USD	-9.23	-3.09	-0.45	2.36	8.71	-8.09	-2.47	0.00	1.49	5.67	-0.47	-0.08	0.02	0.23	0.73	-4.11	-1.01	0.02	1.67	4.23
PTE/USD	-4.84	-2.53	-0.25	1.63	4.40	-4.28	-1.87	-0.31	0.51	2.07	-0.68	-0.26	-0.02	0.15	0.57	-0.83	-0.36	-0.01	0.22	0.92

Notes: This table presents the results of forecasting end-of-month exchange rate 1-month ahead by the no-change prediction, the rolling OLS, recursive OLS, SRidge and EWA methods based on decoupled fundamentals. For each method, the time-series of the differences between the monthly forecasting errors of the no-change predictions and the ones of the method under scrutiny are computed. The table reports the quantiles of these time-series at the 10%, 25%, 50%, 75% and 90% levels, to give an idea of the distributions of the monthly forecasting errors. Distributions of errors that are shifted towards positive values indicate methods that outperfom the no-change prediction. Sample period for the exchange rates: March 1973 – December 2014 unless shorter (indicated in Appendix A).

Table D.15: 1-month ahead forecasts for the PPP, UIRP, monetary model and all fundamentals: shorter-period sample (1980 onwards, decoupled fundamentals).

Currency pair	No change	Rolling regression				Recursive regression				SRidge				EWA			
	RMSE x 100	Theil ratio	CW p-value	DM statistic	DM p-value	Theil ratio	CW p-value	DM statistic	DM p-value	Theil ratio	CW p-value	DM statistic	DM p-value	Theil ratio	CW p-value	DM statistic	DM p-value
PPP fundamental																	
USD/GBP	2.6343	**0.9999**	0.132	0.0073	0.063 *	**0.9981**	0.056 *	0.1405	0.114	**0.9981**	0.078 *	0.8516	0.032 **	**0.9981**	0.202	0.2321	0.261
JPY/USD	3.0508	1.0098	0.718	-1.0566	0.527	1.0036	0.843	-1.0451	0.679	**0.9999**	0.338	0.3785	0.142	**0.9980**	0.125	0.7678	0.154
CHF/USD	3.2016	1.0140	0.891	-1.7297	0.805	1.0036	0.550	-0.5757	0.404	**0.9992**	0.198	0.6061	0.066 *	**0.9951**	0.039 **	1.3309	0.067 *
CAD/USD	2.1689	**0.9979**	0.122	0.2448	0.018 **	**0.9975**	0.071 *	0.9960	0.006 ***	1.0000	0.584	-0.1745	0.203	1.0015	0.990	-1.8477	0.961
SEK/USD	3.3955	1.0067	0.394	-0.6855	0.241	**0.9964**	0.057 *	0.5724	0.029 **	1.0006	0.462	-0.1691	0.212	**0.9951**	0.037 **	1.0407	0.056 *
DNK/USD	3.0006	1.0097	0.729	-1.4442	0.634	1.0004	0.358	-0.1056	0.164	**0.9985**	0.106	0.9276	0.018 **	**0.9918**	0.005 ***	1.9845	0.004 ***
USD/AUD	3.2036	**0.9952**	0.044 **	0.3348	0.028 **	**0.9962**	0.038 **	0.9151	0.015 **	**0.9977**	0.216	0.4713	0.067 *	**0.9977**	0.183	0.6150	0.090 *
FRF/USD	3.0352	1.0136	0.702	-1.1107	0.663	1.0056	0.961	-1.5651	0.863	1.0045	0.865	-1.0074	0.642	**0.9944**	0.123	0.7805	0.112
DEM/USD	3.0482	1.0290	0.978	-1.3175	0.736	1.0147	0.919	-1.2286	0.756	**0.9983**	0.182	0.8222	0.036 **	**0.9897**	0.030 **	1.4106	0.024 **
ITL/USD	3.2141	1.0025	0.370	-0.3244	0.291	**0.9994**	0.301	0.0590	0.198	**0.9986**	0.292	0.2960	0.162	**0.9959**	0.214	0.4274	0.169
NLG/USD	3.0634	1.0143	0.885	-1.3641	0.781	1.0156	0.736	-0.9276	0.605	**0.9978**	0.240	0.5537	0.074 *	**0.9912**	0.050 *	1.2584	0.040 **
PTE/USD	3.1025	1.0164	0.549	-0.9713	0.747	1.0093	0.518	-0.5240	0.618	1.0063	0.483	-0.4045	0.496	**0.9986**	0.262	0.1356	0.064 *
UIRP fundamental																	
USD/GBP	2.6343	1.0400	0.804	-1.4456	0.683	1.0344	0.760	-1.3949	0.797	1.0184	0.688	-0.9752	0.687	1.0025	0.377	-0.2823	0.477
JPY/USD	3.0508	1.0151	0.317	-0.9629	0.380	1.0056	0.134	-0.4047	0.285	1.0004	0.471	-0.1978	0.298	**0.9951**	0.059 *	0.9300	0.142
CHF/USD	3.2016	1.0183	0.307	-1.0100	0.456	1.0170	0.300	-0.8907	0.609	**0.9980**	0.086 *	1.1408	0.024 **	**0.9933**	0.022 **	1.3955	0.027 **
CAD/USD	2.1689	1.0066	0.530	-0.9144	0.337	1.0031	0.401	-0.7283	0.463	1.0000	0.717	-0.5477	0.435	1.0003	0.621	-0.3582	0.453
SEK/USD	3.3927	1.0504	0.886	-1.5555	0.793	1.0298	0.786	-0.9937	0.626	**0.9910**	0.058 *	0.9928	0.031 **	**0.9833**	0.050 *	0.8884	0.123
DNK/USD	3.0006	1.0208	0.902	-1.5845	0.740	1.0068	0.677	-0.8911	0.563	**0.9986**	0.100 *	1.1028	0.020 **	**0.9938**	0.035 **	0.8994	0.079 *
USD/AUD	3.2081	1.0110	0.259	-0.7166	0.221	1.0002	0.366	-0.0692	0.137	**0.9997**	0.366	0.1680	0.137	**0.9986**	0.227	0.3382	0.184
FRF/USD	3.0352	1.0234	0.907	-1.4493	0.819	1.0207	0.745	-0.9752	0.662	1.0001	0.395	-0.0127	0.241	**0.9902**	0.088 *	0.6242	0.129
DEM/USD	3.0482	1.0358	0.742	-1.1143	0.673	1.0226	0.807	-1.0270	0.664	1.0000	0.405	-0.0003	0.261	**0.9853**	0.039 **	1.0974	0.035 **
ITL/USD	3.2141	1.0044	0.283	-0.1777	0.258	1.0175	0.771	-1.0810	0.731	**0.9909**	0.111	0.7981	0.099 *	**0.9789**	0.031 **	0.9525	0.061 *
NLG/USD	3.0634	1.0884	0.678	-1.1892	0.732	1.0524	0.646	-1.0759	0.735	**0.9961**	0.177	0.6878	0.094 *	**0.9877**	0.063 *	0.9017	0.061 *
PTE/USD																	
Monetary model fundamentals																	
USD/GBP	2.6343	1.0335	0.752	-2.3566	0.829	1.0154	0.615	-1.3315	0.584	**0.9900**	0.033 **	0.6315	0.035 **	1.0058	0.310	-0.4996	0.348
JPY/USD	3.0536	1.0216	0.234	-1.3911	0.406	1.0180	0.296	-1.2058	0.588	1.0058	0.287	-0.5457	0.423	**0.9999**	0.032 **	0.0091	0.186
CHF/USD	3.2169	1.0305	0.644	-1.6186	0.515	1.0137	0.598	-1.3470	0.640	1.0046	0.613	-0.9298	0.596	1.0091	0.414	-0.8724	0.577
CAD/USD	2.1708	1.0395	0.472	-1.4050	0.299	1.0100	0.662	-0.8886	0.374	1.0131	0.827	-0.9607	0.568	1.0090	0.919	-1.4540	0.791
SEK/USD	3.3927	1.0467	0.941	-2.4612	0.872	1.0195	0.888	-2.0119	0.887	1.0087	0.376	-0.4347	0.307	1.0057	0.333	-0.5362	0.513
DNK/USD	3.0252	1.0407	0.979	-2.3514	0.879	1.0087	0.944	-1.7572	0.845	1.0010	0.740	-0.6841	0.459	1.0062	0.499	-0.7246	0.449
USD/AUD	3.2191	1.0277	0.512	-1.3930	0.351	1.0109	0.952	-2.1154	0.922	1.0014	0.612	-0.3317	0.253	1.0028	0.426	-0.2994	0.269
FRF/USD	3.0352	1.0488	0.712	-1.6674	0.842	1.0453	0.810	-1.7522	0.904	1.0006	0.399	-0.0951	0.260	1.0116	0.530	-0.7274	0.537
DEM/USD	3.0482	1.0220	0.370	-0.7846	0.424	1.0132	0.274	-0.5195	0.381	**0.9993**	0.109	0.0907	0.221	1.0019	0.156	-0.0993	0.217
ITL/USD	3.2141	1.0179	0.381	-0.6209	0.310	1.0189	0.467	-0.7490	0.435	**0.9937**	0.148	0.5077	0.081 *	1.0004	0.307	-0.0211	0.362
NLG/USD	3.0634	1.0459	0.771	-1.4259	0.874	1.0076	0.514	-0.5150	0.623	**0.9948**	0.140	0.7099	0.111	1.0012	0.308	-0.0773	0.215
PTE/USD																	
All fundamentals																	
USD/GBP	2.6343	1.0881	0.287	-2.9954	0.588	1.0389	0.592	-1.4756	0.341	**0.9934**	0.039 **	0.8314	0.027 **	1.0050	0.368	-0.5108	0.425
JPY/USD	3.0536	1.0604	0.183	-2.0425	0.159	1.0281	0.164	-1.4731	0.362	1.0029	0.332	-0.3908	0.368	**0.9978**	0.020 **	0.2164	0.202
CHF/USD	3.2169	1.1053	0.678	-2.6538	0.518	1.0301	0.313	-1.6970	0.543	1.0034	0.602	-0.7146	0.527	1.0093	0.402	-0.8452	0.682
CAD/USD	2.1708	1.0377	0.164	-1.5192	0.021 **	1.0090	0.318	-0.5139	0.033 **	1.0035	0.845	-1.0152	0.640	1.0023	0.931	-1.5095	0.879
SEK/USD	3.3927	1.1693	0.893	-1.6888	0.273	1.1165	0.823	-1.1405	0.324	1.0104	0.906	-1.3627	0.841	**0.9895**	0.044 **	0.5311	0.226
DNK/USD	3.0252	1.0763	0.669	-2.8720	0.592	1.0293	0.631	-1.5507	0.390	**0.9999**	0.389	0.0609	0.180	1.0055	0.523	-0.7125	0.518
USD/AUD	3.2191	1.0405	0.087 *	-1.3170	0.027 **	1.0224	0.731	-1.5560	0.417	1.0002	0.441	-0.0495	0.182	**0.9981**	0.130	0.1510	0.175
FRF/USD	3.0352	1.1011	0.595	-1.8930	0.705	1.0475	0.643	-1.3648	0.638	**0.9992**	0.336	0.1564	0.165	1.0127	0.420	-0.6667	0.556
DEM/USD	3.0482	1.0721	0.571	-1.3458	0.295	1.0496	0.591	-1.1094	0.359	**0.9948**	0.081 *	0.7301	0.064 *	**0.9938**	0.100 *	0.3100	0.182
ITL/USD	3.2141	**0.9967**	0.063 *	0.0655	0.030 **	1.0259	0.185	-0.6482	0.299	**0.9911**	0.134	0.7764	0.056 *	**0.9905**	0.126	0.4129	0.170
NLG/USD	3.0634	1.0656	0.348	-1.5904	0.719	1.0579	0.760	-1.7240	0.878	**0.9947**	0.138	0.7694	0.108	**0.9942**	0.152	0.3015	0.179
PTE/USD																	

Notes: This table presents the results of forecasting end-of-month exchange rate 1-month ahead by the no-change prediction, the rolling OLS, recursive OLS, SRidge and EWA methods for decoupled fundamentals for the time period 1980–2014. Column 1 shows the RMSE values for the no-change prediction. For each method, the Theil ratio (RMSE of the given method / RMSE of the no-change prediction), the CW-p-values, the DM statistic and its bootstrapped p-value are shown. The CW and DM tests are one-sided tests of equal out-of-sample prediction accuracy (H_0) against superior out-of-sample prediction accuracy (H_1) for the methods considered compared to the no-change prediction. ***, **, and * denote statistical significance at the 1 %, 5 %, and 10 % levels. Theil ratios <1 are indicated in bold.

Table D.16: Economic criterion for evaluating forecasts for the EWA algorithm without the DNK/USD exchange rate.

Portfolio / c constraint	Annualized: Returns (in %): No change (with c constraint)	Returns (in %): Portfolio	Performance fee of portfolio w.r.t. no change (in bps)	Premium return of portfolio w.r.t. no change (in bps)	Sharpe ratios: No change (with c constraint)	Sharpe ratios: Portfolio	Sortino ratios: No change (with c constraint)	Sortino ratios: Portfolio
PPP fundamental								
EWA / c = 0.5	10.36	11.38	146	177	0.60	0.75	0.63	1.06
EWA / c = 1	11.43	12.83	241	289	0.61	0.81	0.63	1.04
EWA / c = 2	10.87	12.36	226	256	0.56	0.75	0.61	0.94
EWA / c = 5	11.11	12.33	193	223	0.58	0.74	0.62	0.90
EWA / c = 10	11.11	12.33	193	223	0.58	0.74	0.62	0.90
EWA / no c	11.11	12.33	193	223	0.58	0.74	0.62	0.90
UIRP fundamental								
EWA / c = 0.5	9.92	11.84	246	288	0.57	0.82	0.60	1.38
EWA / c = 1	11.04	13.32	326	409	0.58	0.86	0.60	1.51
EWA / c = 2	10.49	12.70	324	403	0.54	0.82	0.59	1.57
EWA / c = 5	10.73	12.26	276	361	0.55	0.79	0.60	1.48
EWA / c = 10	10.73	12.26	276	361	0.55	0.79	0.60	1.48
EWA / no c	10.73	12.26	276	361	0.55	0.79	0.60	1.48
Monetary models fundamentals								
EWA / c = 0.5	10.33	9.38	-36	-8	0.59	0.55	0.62	0.72
EWA / c = 1	11.40	10.75	73	121	0.60	0.64	0.62	0.81
EWA / c = 2	10.85	9.70	12	41	0.56	0.54	0.60	0.67
EWA / c = 5	11.08	9.82	-2	27	0.57	0.54	0.61	0.66
EWA / c = 10	11.08	9.82	-2	27	0.57	0.54	0.61	0.66
EWA / no c	11.08	9.82	-2	27	0.57	0.54	0.61	0.66
All fundamentals								
EWA / c = 0.5	10.33	9.71	20	61	0.59	0.61	0.62	1.07
EWA / c = 1	11.40	10.61	59	133	0.60	0.63	0.62	1.03
EWA / c = 2	10.85	10.47	75	153	0.56	0.60	0.60	1.04
EWA / c = 5	11.08	10.27	38	128	0.57	0.58	0.61	1.00
EWA / c = 10	11.08	10.27	38	128	0.57	0.58	0.61	1.00
EWA / no c	11.08	10.27	38	128	0.57	0.58	0.61	1.00

Notes: This table presents the comparison of performance of portfolios formed using forecasts from the EWA algorithm and the no-change prediction based on decoupled fundamentals without the DNK/USD exchange rate, using the procedure described in Section 5.4. Column 1 gives the conditions under which the portfolios were constructed. In columns 2 and 3 the annualized returns are given for the EWA and no-change prediction-based portfolios respectively. Column 4 shows the performance fee in basis points (bps) while column 5 the premium return (in bps) of the EWA-based portfolios relative to the ones formed based on the no-change forecast. The Sharpe and Sortino ratios for portfolios created using the two competing forecasting methods are reported in columns 6-10. For details on how these portfolios were created please consult Appendix C.3. The results for the portfolios based on the no-change exchange rate predictions vary because of the constraints on weights and because for different sets of fundamentals different time periods are considered depending on the availability of data.

Table D.17: Economic criterion for evaluating forecasts for the EWA algorithm with an additional summation-to-less-than-1 constraint on the weights.

Portfolio / c constraint	Annualized Returns (in %) No change (with c constraint)	Returns (in %) Portfolio	Performance fee of portfolio w.r.t. no change (in bps)	Premium return of portfolio w.r.t. no change (in bps)	Sharpe ratios No change (with c constraint)	Sharpe ratios Portfolio	Sortino ratios No change (with c constraint)	Sortino ratios Portfolio
PPP fundamental								
EWA / c = 0.5	10.76	11.77	155	185	0.64	0.80	0.67	1.13
EWA / c = 1	11.55	13.07	244	301	0.61	0.82	0.63	1.08
EWA / c = 2	11.58	13.54	276	325	0.61	0.84	0.63	1.05
EWA / c = 5	11.72	13.35	231	278	0.63	0.82	0.67	1.06
EWA / c = 10	11.69	13.39	236	284	0.63	0.83	0.67	1.08
EWA / no c	11.69	13.39	236	284	0.63	0.83	0.67	1.08
UIRP fundamental								
EWA / c = 0.5	10.32	11.89	196	239	0.60	0.79	0.64	1.29
EWA / c = 1	11.16	13.02	279	365	0.59	0.82	0.61	1.34
EWA / c = 2	11.19	12.99	283	375	0.59	0.82	0.61	1.43
EWA / c = 5	11.33	12.14	197	275	0.60	0.77	0.65	1.35
EWA / c = 10	11.30	12.11	204	280	0.60	0.78	0.65	1.38
EWA / no c	11.30	12.11	204	280	0.60	0.78	0.65	1.38
Monetary models fundamentals								
EWA / c = 0.5	10.70	8.54	-117	-87	0.63	0.50	0.67	0.67
EWA / c = 1	11.46	9.34	-56	6	0.60	0.52	0.63	0.68
EWA / c = 2	11.51	8.89	-114	-50	0.60	0.47	0.63	0.62
EWA / c = 5	11.59	8.46	-168	-101	0.62	0.43	0.66	0.62
EWA / c = 10	11.55	8.41	-175	-108	0.62	0.43	0.66	0.61
EWA / no c	11.55	8.41	-175	-108	0.62	0.43	0.66	0.61
All fundamentals								
EWA / c = 0.5	10.70	8.52	-100	-63	0.63	0.51	0.67	0.84
EWA / c = 1	11.46	9.56	-12	62	0.60	0.56	0.63	0.91
EWA / c = 2	11.51	9.20	-81	3	0.60	0.50	0.63	0.89
EWA / c = 5	11.59	8.57	-166	-82	0.62	0.44	0.66	0.80
EWA / c = 10	11.55	8.48	-175	-91	0.62	0.43	0.66	0.79
EWA / no c	11.55	8.48	-175	-91	0.62	0.43	0.66	0.79

Notes: This table presents the comparison of performance of portfolios formed using forecasts from the EWA algorithm and the no-change prediction based on decoupled fundamentals with an additional summation-to-less-than-1 constraint on the weights, using the procedure described in Section 5.4. Column 1 gives the conditions under which the portfolios were constructed. In columns 2 and 3 the annualized returns are given for the EWA and no-change prediction-based portfolios respectively. Column 4 shows the performance fee in basis points (bps) while column 5 the premium return (in bps) of the EWA-based portfolios relative to the ones formed based on the no-change forecast. The Sharpe and Sortino ratios for portfolios created using the two competing forecasting methods are reported in columns 6-10. For details on how these portfolios were created please consult Appendix C.3. The results for the portfolios based on the no-change exchange rate predictions vary because of the constraints on weights and because for different sets of fundamentals different time periods are considered depending on the availability of data.

Table D.18: Economic criterion for evaluating forecasts for the EWA algorithm without the DNK/USD exchange rate but with an additional summation-to-less-than-1 constraint on the weights.

Portfolio / c constraint	Annualized							
	Returns (in %)		Performance fee of portfolio w.r.t. no change (in bps)	Premium return of portfolio w.r.t. no change (in bps)	Sharpe ratios		Sortino ratios	
	No change (with c constraint)	Portfolio			No change (with c constraint)	Portfolio	No change (with c constraint)	Portfolio
PPP fundamental								
EWA / c = 0.5	10.24	11.30	158	185	0.60	0.76	0.64	1.08
EWA / c = 1	10.89	12.71	288	340	0.56	0.80	0.58	1.05
EWA / c = 2	10.57	12.28	252	282	0.54	0.74	0.58	0.94
EWA / c = 5	10.73	12.20	223	253	0.54	0.72	0.59	0.89
EWA / c = 10	10.73	12.20	223	253	0.54	0.72	0.59	0.89
EWA / no c	10.73	12.20	223	253	0.54	0.72	0.59	0.89
UIRP fundamental								
EWA / c = 0.5	9.81	11.66	237	275	0.56	0.81	0.60	1.37
EWA / c = 1	10.51	13.17	367	452	0.53	0.84	0.56	1.48
EWA / c = 2	10.19	12.49	338	418	0.51	0.80	0.56	1.53
EWA / c = 5	10.35	12.13	304	391	0.52	0.77	0.57	1.46
EWA / c = 10	10.35	12.13	304	391	0.52	0.77	0.57	1.46
EWA / no c	10.35	12.13	304	391	0.52	0.77	0.57	1.46
Monetary models fundamentals								
EWA / c = 0.5	10.22	9.09	-35	-9	0.59	0.55	0.63	0.74
EWA / c = 1	10.86	10.61	117	168	0.55	0.63	0.58	0.80
EWA / c = 2	10.54	9.68	48	77	0.53	0.54	0.58	0.67
EWA / c = 5	10.70	9.68	31	61	0.54	0.53	0.58	0.65
EWA / c = 10	10.70	9.68	31	61	0.54	0.53	0.58	0.65
EWA / no c	10.70	9.68	31	61	0.54	0.53	0.58	0.65
All fundamentals								
EWA / c = 0.5	10.22	9.26	-11	28	0.59	0.58	0.63	1.02
EWA / c = 1	10.86	10.39	93	171	0.55	0.61	0.58	1.02
EWA / c = 2	10.54	10.28	90	168	0.53	0.58	0.58	1.01
EWA / c = 5	10.70	10.12	64	156	0.54	0.57	0.58	0.98
EWA / c = 10	10.70	10.12	64	156	0.54	0.57	0.58	0.98
EWA / no c	10.70	10.12	64	156	0.54	0.57	0.58	0.98

This table presents the comparison of performance of portfolios formed using forecasts from the EWA algorithm and the no-change prediction based on decoupled fundamentals with an additional summation-to-less-than-1 constraint on the weights and without the DNK/USD exchange rate, using the procedure described in Section 5.4. Column 1 gives the conditions under which the portfolios were constructed. In columns 2 and 3 the annualized returns are given for the EWA and no-change prediction-based portfolios respectively. Column 4 shows the performance fee in basis points (bps) while column 5 the premium return (in bps) of the EWA-based portfolios relative to the ones formed based on the no-change forecast. The Sharpe and Sortino ratios for portfolios created using the two competing forecasting methods are reported in columns 6-10. For details on how these portfolios were created please consult Appendix C.3. The results for the portfolios based on the no-change exchange rate predictions vary because of the constraints on weights and because for different sets of fundamentals different time periods are considered depending on the availability of data.

Table D.19: 1-month ahead forecasts for the PPP, UIRP, monetary model and all fundamentals based on decoupled fundamentals, for a sample February 1999 – December 2014.

Currency pair	No change	Rolling regression				Recursive regression				SRidge				EWA			
	RMSE x 100	Theil ratio	CW p-value	DM statistic	DM p-value	Theil ratio	CW p-value	DM statistic	DM p-value	Theil ratio	CW p-value	DM statistic	DM p-value	Theil ratio	CW p-value	DM statistic	DM p-value
PPP fundamental																	
USD/GBP	2.5258	1.0085	0.170	-0.5437	0.161	1.0068	0.286	-0.6949	0.357	**0.9970**	0.259	0.1330	0.101	1.0016	0.605	-0.2454	0.412
JPY/USD	2.7610	1.0223	0.543	-0.8540	0.353	1.0139	0.826	-0.9946	0.596	**0.9975**	0.150	0.5566	0.065 *	**0.9981**	0.179	0.6756	0.086 *
CHF/USD	3.1193	1.0376	0.920	-1.8890	0.873	1.0247	0.933	-2.0651	0.970	1.0082	0.673	-0.8226	0.568	**0.9999**	0.410	0.0222	0.612
CAD/USD	2.7942	1.0126	0.363	-0.6230	0.212	1.0005	0.276	-0.0411	0.104	1.0084	0.732	-1.0721	0.702	1.0048	0.882	-1.2080	0.859
SEK/USD	3.5100	1.0245	0.295	-1.1734	0.488	1.0137	0.398	-0.6253	0.357	**0.9939**	0.206	0.2098	0.086 *	**0.9950**	0.104	0.7234	0.108
DNK/USD	3.0702	1.0356	0.710	-0.8647	0.358	1.0052	0.448	-0.3771	0.243	1.0016	0.289	-0.0905	0.178	**0.9972**	0.224	0.3895	0.276
USD/AUD	3.8442	**0.9993**	0.083 *	0.0284	0.139	**0.9866**	0.041 **	0.7394	0.069 *	**0.9999**	0.301	0.0030	0.282	1.0017	0.561	-0.2665	0.461
UIRP fundamental																	
USD/GBP	2.5258	1.0115	0.385	-0.5488	0.134	1.0202	0.910	-1.9441	0.945	**0.9979**	0.134	0.9643	0.023 **	**0.9972**	0.034 **	1.0874	0.079 *
JPY/USD	2.7610	1.0630	0.747	-1.0644	0.353	1.0290	0.618	-1.3238	0.691	1.0053	0.947	-1.4617	0.902	**0.9997**	0.361	0.2149	0.257
CHF/USD	3.1193	1.0229	0.408	-1.1126	0.460	1.0100	0.582	-0.7324	0.416	1.0007	0.538	-0.2525	0.285	**0.9988**	0.259	0.2832	0.323
CAD/USD	2.7942	1.0182	0.681	-1.4171	0.594	1.0100	0.878	-1.3278	0.709	1.0067	0.828	-0.9738	0.573	1.0012	0.703	-0.5686	0.590
SEK/USD	3.5047	1.0266	0.509	-1.1740	0.439	1.0278	0.952	-1.5156	0.808	**0.9999**	0.305	0.0125	0.147	**0.9972**	0.215	0.3806	0.252
DNK/USD	3.0702	1.0354	0.713	-1.0895	0.349	1.0338	0.908	-1.2070	0.620	1.0045	0.609	-0.4274	0.276	**0.9954**	0.154	0.6001	0.169
USD/AUD	3.8257	1.0417	0.943	-2.0130	0.867	1.0326	0.939	-1.4754	0.814	1.0071	0.621	-1.1503	0.733	**0.9996**	0.413	0.0782	0.296
Monetary model fundamentals																	
USD/GBP	2.5258	1.1494	0.177	-1.3749	0.379	1.0517	0.926	-1.7443	0.809	**0.9982**	0.067 *	0.0595	0.095 *	**0.9995**	0.312	0.0349	0.165
JPY/USD	2.7659	1.0786	0.409	-1.4662	0.318	1.0320	0.620	-1.1796	0.426	1.0153	0.737	-1.0185	0.581	1.0098	0.509	-0.6569	0.445
CHF/USD	3.1525	1.0950	0.777	-1.4593	0.538	1.0503	0.898	-1.5718	0.857	1.0024	0.787	-0.8010	0.642	1.0125	0.825	-1.1172	0.876
CAD/USD	2.8019	1.1202	0.255	-1.3916	0.328	1.0432	0.399	-1.0928	0.450	1.0355	0.781	-1.0937	0.670	1.0110	0.847	-1.1910	0.798
SEK/USD	3.5047	1.1182	0.632	-1.5861	0.456	1.0672	0.899	-2.0926	0.919	1.0456	0.867	-1.0169	0.623	1.0171	0.883	-1.1029	0.825
DNK/USD	3.1342	1.1042	0.471	-1.4754	0.539	1.0352	0.774	-1.7878	0.881	1.0090	0.694	-0.8862	0.656	1.0152	0.822	-1.1638	0.781
USD/AUD	3.8947	1.1123	0.256	-1.4525	0.519	1.0444	0.637	-1.5648	0.781	1.0047	0.460	-0.3709	0.377	1.0038	0.386	-0.2533	0.502
All fundamentals																	
USD/GBP	2.5258	1.1026	0.016 **	-1.0425	0.039 **	1.0296	0.191	-1.0780	0.245	**0.9932**	0.080 *	0.2118	0.066 *	**0.9995**	0.306	0.0464	0.201
JPY/USD	2.7659	1.1423	0.059 *	-1.9922	0.186	1.1092	0.711	-1.9643	0.709	1.0143	0.768	-0.9702	0.535	1.0109	0.580	-0.7225	0.535
CHF/USD	3.1525	1.3213	0.852	-1.9938	0.356	1.1486	0.969	-2.4218	0.914	1.0009	0.678	-0.5389	0.527	1.0059	0.885	-1.2726	0.923
CAD/USD	2.8019	1.2338	0.430	-1.6283	0.105	1.0547	0.395	-1.7072	0.578	1.0217	0.759	-1.0179	0.637	1.0100	0.879	-1.2133	0.842
SEK/USD	3.5047	1.1306	0.369	-1.3005	0.041 **	1.0977	0.452	-1.4593	0.355	1.0322	0.621	-0.7284	0.448	1.0110	0.513	-0.5562	0.618
DNK/USD	3.1342	1.2111	0.540	-2.8525	0.709	1.0568	0.398	-1.1348	0.218	1.0149	0.736	-0.9102	0.645	1.0124	0.746	-0.8549	0.683
USD/AUD	3.8947	1.2327	0.205	-1.5940	0.101	1.0391	0.134	-0.8867	0.149	1.0063	0.446	-0.4445	0.428	1.0045	0.395	-0.2805	0.464

Notes: This table presents the results of forecasting end-of-month exchange rate 1-month ahead by the no-change prediction, the rolling OLS, recursive OLS, SRidge and EWA methods for decoupled fundamentals on a sample February 1999 – December 2014. Column 1 shows the RMSE values for the no-change prediction. For each method, the Theil ratio (RMSE of the given method / RMSE of the no-change prediction), the CW-p-values, the DM statistic and its bootstrapped p-value are shown. The CW and DM tests are one-sided tests of equal out-of-sample prediction accuracy (H_0) against superior out-of-sample prediction accuracy (H_1) for the methods considered compared to the no-change prediction. ***, **, and * denote statistical significance at the 1 %, 5 %, and 10 % levels. Theil ratios < 1 are indicated in bold.

Table D.20: 1-month ahead forecasts on the real-time data set for the PPP, UIRP, monetary model and all fundamentals based on decoupled fundamentals, for a sample February 1999 – December 2014.

Currency pair	No change	Rolling regression				Recursive regression				SRidge				EWA			
	RMSE x 100	Theil ratio	CW p-value	DM statistic	DM p-value	Theil ratio	CW p-value	DM statistic	DM p-value	Theil ratio	CW p-value	DM statistic	DM p-value	Theil ratio	CW p-value	DM statistic	DM p-value
PPP fundamental																	
USD/GBP	2.5258	1.0002	0.070 *	-0.0118	0.060 *	**0.9957**	0.055 *	0.2563	0.068 *	**0.9881**	0.134	0.4070	0.062 *	**0.9996**	0.401	0.0751	0.277
JPY/USD	2.7610	1.0188	0.482	-0.8736	0.351	1.0141	0.823	-1.3547	0.771	**0.9935**	0.093 *	0.6978	0.040 **	**0.9956**	0.069 *	0.9662	0.051 *
CHF/USD	3.1193	1.0403	0.863	-1.5331	0.689	1.0208	0.921	-1.6044	0.858	1.0127	0.724	-0.8081	0.519	1.0015	0.530	-0.2597	0.636
CAD/USD	2.7942	1.0048	0.166	-0.1970	0.096 *	**0.9932**	0.116	0.4592	0.044 **	1.0085	0.722	-1.1197	0.735	1.0057	0.896	-1.1590	0.820
SEK/USD	3.5100	1.0269	0.333	-1.3824	0.632	1.0139	0.468	-0.6405	0.379	**0.9986**	0.214	0.0478	0.159	**0.9960**	0.147	0.6227	0.080 *
DNK/USD	3.0702	1.0343	0.782	-0.9913	0.422	1.0092	0.840	-0.9107	0.543	1.0070	0.401	-0.3545	0.279	**0.9988**	0.319	0.1613	0.394
USD/AUD	3.8442	1.0052	0.150	-0.2499	0.248	**0.9867**	0.045 **	0.7369	0.088 *	1.0113	0.396	-0.9772	0.739	1.0024	0.623	-0.3818	0.484
UIRP fundamental																	
USD/GBP	2.5258	1.0117	0.387	-0.5560	0.135	1.0202	0.910	-1.9439	0.926	**0.9979**	0.134	0.9643	0.017 **	**0.9972**	0.034 **	1.0863	0.066 *
JPY/USD	2.7610	1.0630	0.747	-1.0644	0.378	1.0290	0.618	-1.3241	0.726	1.0053	0.947	-1.4627	0.914	**0.9997**	0.361	0.2137	0.288
CHF/USD	3.1193	1.0229	0.407	-1.1134	0.467	1.0100	0.581	-0.7307	0.455	1.0007	0.538	-0.2520	0.319	**0.9988**	0.259	0.2838	0.344
CAD/USD	2.7942	1.0183	0.681	-1.4186	0.614	1.0100	0.878	-1.3279	0.729	1.0067	0.828	-0.9738	0.614	1.0012	0.703	-0.5674	0.597
SEK/USD	3.5047	1.0266	0.509	-1.1752	0.479	1.0278	0.952	-1.5146	0.828	**0.9999**	0.305	0.0122	0.185	**0.9972**	0.215	0.3808	0.271
DNK/USD	3.0702	1.0354	0.713	-1.0899	0.355	1.0338	0.908	-1.2061	0.622	1.0045	0.609	-0.4273	0.280	**0.9954**	0.154	0.6002	0.176
USD/AUD	3.8257	1.0416	0.943	-2.0129	0.874	1.0326	0.939	-1.4749	0.810	1.0071	0.621	-1.1503	0.748	**0.9996**	0.413	0.0777	0.318
Monetary model fundamentals																	
USD/GBP	2.5258	1.1412	0.242	-1.2863	0.222	1.0432	0.980	-2.6339	0.980	**0.9847**	0.048 **	0.5622	0.038 **	**0.9932**	0.136	0.4358	0.085 *
JPY/USD	2.7659	1.0752	0.243	-1.2117	0.255	1.0139	0.154	-0.4874	0.166	1.0127	0.794	-1.1954	0.718	1.0003	0.179	-0.0386	0.239
CHF/USD	3.1525	1.0634	0.896	-1.4040	0.423	1.0169	0.652	-1.1913	0.632	1.0048	0.921	-1.4640	0.905	1.0134	0.912	-1.5559	0.908
CAD/USD	2.8019	1.1077	0.966	-1.6715	0.507	1.0336	0.972	-1.5207	0.728	1.0238	0.705	-0.7849	0.502	1.0094	0.803	-0.9744	0.752
SEK/USD	3.5047	1.1312	0.656	-1.5253	0.403	1.0434	0.904	-1.6997	0.767	1.0253	0.642	-0.7035	0.411	**0.9916**	0.126	0.4394	0.100 *
DNK/USD	3.1342	1.0795	0.843	-1.4930	0.552	1.0255	0.441	-0.9238	0.540	1.0131	0.776	-0.6939	0.511	1.0180	0.657	-0.8030	0.708
USD/AUD	3.8947	1.1073	0.621	-1.9913	0.777	1.0326	0.365	-0.9959	0.520	1.0064	0.211	-0.3726	0.322	**0.9918**	0.183	0.4194	0.128
All fundamentals																	
USD/GBP	2.5258	1.0983	0.185	-1.3071	0.045 **	1.0160	0.042 **	-0.2801	0.017 **	**0.9838**	0.046 **	0.6158	0.026 **	**0.9897**	0.087 *	0.6106	0.071 *
JPY/USD	2.7659	1.1896	0.440	-2.0695	0.207	1.0928	0.792	-1.5691	0.481	1.0102	0.719	-0.9773	0.563	1.0018	0.263	-0.1755	0.305
CHF/USD	3.1525	1.3392	0.764	-1.8914	0.264	1.1393	0.889	-2.5819	0.908	1.0056	0.905	-1.2996	0.851	1.0109	0.865	-1.3334	0.877
CAD/USD	2.8019	1.1503	0.731	-2.3459	0.401	1.0570	0.197	-1.7003	0.574	1.0246	0.777	-0.9360	0.585	1.0096	0.807	-0.9836	0.765
SEK/USD	3.5047	1.1584	0.490	-1.3902	0.050 **	1.1025	0.649	-2.1765	0.769	1.0209	0.547	-0.5658	0.337	**0.9921**	0.138	0.4162	0.099 *
DNK/USD	3.1342	1.1709	0.480	-2.7254	0.536	1.1167	0.885	-1.9997	0.617	1.0155	0.826	-0.7561	0.555	1.0114	0.455	-0.5986	0.668
USD/AUD	3.8947	1.2219	0.318	-2.2326	0.420	1.0677	0.274	-1.6388	0.563	1.0054	0.229	-0.3417	0.453	**0.9929**	0.196	0.4334	0.207

Notes: This table presents the results of forecasting end-of-month exchange rate 1-month ahead by the no-change prediction, the rolling OLS, recursive OLS, SRidge and EWA methods for decoupled fundamentals on a sample February 1999 – December 2014 using real time data. Column 1 shows the RMSE values for the no-change prediction. For each method, the Theil ratio (RMSE of the given method / RMSE of the no-change prediction), the CW-p-values, the DM statistic and its bootstrapped p-value are shown. The CW and DM tests are one-sided tests of equal out-of-sample prediction accuracy (H_0) against superior out-of-sample prediction accuracy (H_1) for the methods considered compared to the no-change prediction. ***, **, and * denote statistical significance at the 1 %, 5 %, and 10 % levels. Theil ratios < 1 are indicated in bold.

Table D.21: Relative forecasting performance of the machine learning methods vs. rolling and recursive regressions based on decoupled fundamentals, for a sample February 1999 – December 2014.

Currency pair	SRidge vs.						EWA vs.					
	Rolling regression			Recursive regression			Rolling regression			Recursive regression		
	Ratio of RMSEs	DM statistic	DM p-value	Ratio of RMSEs	DM statistic	DM p-value	Ratio of RMSEs	DM statistic	DM p-value	Ratio of RMSEs	DM statistic	DM p-value
PPP fundamental												
USD/GBP	**0.9886**	0.5362	0.787	**0.9902**	0.3986	0.663	**0.9932**	0.4560	0.860	**0.9948**	0.4730	0.709
JPY/USD	**0.9758**	1.0566	0.487	**0.9838**	1.3024	0.181	**0.9763**	0.9944	0.552	**0.9843**	1.2206	0.227
CHF/USD	**0.9717**	1.9816	0.084 *	**0.9839**	1.8924	0.038 **	**0.9637**	2.2734	0.049 **	**0.9758**	2.2727	0.017 **
CAD/USD	**0.9959**	0.2305	0.863	1.0079	-0.6985	0.972	**0.9923**	0.4035	0.874	1.0043	-0.3924	0.962
SEK/USD	**0.9701**	0.7074	0.682	**0.9805**	0.4545	0.600	**0.9712**	1.1070	0.548	**0.9816**	0.6751	0.618
DNK/USD	**0.9672**	0.7902	0.622	**0.9964**	0.1748	0.737	**0.9629**	0.8799	0.656	**0.9920**	0.4564	0.730
USD/AUD	1.0006	-0.0182	0.867	1.0135	-0.5533	0.877	1.0024	-0.0859	0.901	1.0153	-0.8647	0.952
UIRP fundamental												
USD/GBP	**0.9865**	0.7149	0.765	**0.9781**	2.1247	0.022 **	**0.9858**	0.7432	0.789	**0.9774**	1.9211	0.042 **
JPY/USD	**0.9457**	0.9793	0.643	**0.9770**	1.1324	0.295	**0.9405**	1.0830	0.630	**0.9715**	1.3196	0.275
CHF/USD	**0.9783**	1.1929	0.458	**0.9908**	0.7168	0.523	**0.9765**	1.2336	0.469	**0.9889**	0.8088	0.551
CAD/USD	**0.9887**	0.8970	0.652	**0.9968**	0.3985	0.682	**0.9833**	1.3706	0.416	**0.9913**	1.1748	0.328
SEK/USD	**0.9740**	1.5061	0.324	**0.9728**	1.1928	0.281	**0.9714**	1.3757	0.444	**0.9702**	1.2762	0.289
DNK/USD	**0.9702**	1.1742	0.525	**0.9717**	1.2512	0.232	**0.9614**	1.2200	0.570	**0.9629**	1.2641	0.319
USD/AUD	**0.9668**	1.6117	0.255	**0.9752**	1.1770	0.268	**0.9596**	1.8536	0.163	**0.9680**	1.2611	0.256
Monetary model fundamentals												
USD/GBP	**0.8685**	1.2116	0.668	**0.9492**	1.0932	0.447	**0.8696**	1.2522	0.607	**0.9504**	1.3331	0.276
JPY/USD	**0.9414**	1.2863	0.738	**0.9839**	0.7908	0.718	**0.9363**	1.4744	0.571	**0.9785**	1.1625	0.416
CHF/USD	**0.9155**	1.4438	0.524	**0.9544**	1.5701	0.164	**0.9247**	1.3700	0.555	**0.9640**	1.5093	0.180
CAD/USD	**0.9244**	1.3627	0.690	**0.9926**	0.1991	0.921	**0.9025**	1.3600	0.644	**0.9692**	0.9032	0.575
SEK/USD	**0.9351**	1.4602	0.596	**0.9798**	0.7115	0.693	**0.9096**	1.5377	0.568	**0.9531**	2.1666	0.048 **
DNK/USD	**0.9138**	1.3720	0.551	**0.9747**	1.8077	0.106	**0.9194**	1.2076	0.528	**0.9807**	1.0238	0.329
USD/AUD	**0.9033**	1.4026	0.535	**0.9620**	1.3481	0.279	**0.9025**	1.4129	0.540	**0.9611**	1.4685	0.214
All fundamentals												
USD/GBP	**0.9007**	1.3065	0.916	**0.9646**	1.4744	0.515	**0.9065**	1.1263	0.944	**0.9707**	1.2368	0.646
JPY/USD	**0.8879**	2.0237	0.798	**0.9144**	1.9609	0.280	**0.8850**	2.1389	0.738	**0.9114**	2.0812	0.190
CHF/USD	**0.7575**	1.9883	0.668	**0.8714**	2.4017	0.090 *	**0.7613**	1.9973	0.661	**0.8758**	2.4444	0.080 *
CAD/USD	**0.8281**	1.6460	0.899	**0.9687**	1.1890	0.717	**0.8186**	1.6342	0.890	**0.9576**	1.5327	0.505
SEK/USD	**0.9129**	1.3903	0.946	**0.9403**	1.6704	0.497	**0.8942**	1.2641	0.967	**0.9211**	1.3802	0.688
DNK/USD	**0.8380**	2.9586	0.256	**0.9603**	0.8985	0.890	**0.8359**	2.9593	0.243	**0.9579**	0.9716	0.826
USD/AUD	**0.8164**	1.5585	0.929	**0.9684**	0.7568	0.911	**0.8149**	1.5651	0.926	**0.9667**	0.8167	0.888

Notes: This table presents the comparison of forecasting performance between, respectively, SRidge and EWA vs. the rolling and recursive OLS for decoupled fundamentals for the period February 1999–December 2014. Columns 1-6 show the comparison between SRidge and OLS methods, while columns 7-12 for EWA vs. OLS. For each comparison, the Theil ratio (RMSE of the given method / RMSE of the compared) and the DM statistic and its bootstrapped p-value are shown. The DM test is a one-sided test of equal out-of-sample prediction accuracy (H_0) against superior out-of-sample prediction accuracy (H_1) of the machine-learning method considered against the OLS methods. ***, **, and * denote statistical significance at the 1 %, 5 %, and 10 % levels. Theil ratios <1 are indicated in bold.

Table D.22: Relative forecasting performance on the real-time data set (February 1999 – December 2014) of the machine learning methods vs. rolling and recursive regressions based on decoupled fundamentals.

Currency pair	SRidge vs.						EWA vs.					
	Rolling regression			Recursive regression			Rolling regression			Recursive regression		
	Ratio of RMSEs	DM statistic	DM p-value	Ratio of RMSEs	DM statistic	DM p-value	Ratio of RMSEs	DM statistic	DM p-value	Ratio of RMSEs	DM statistic	DM p-value
PPP fundamental												
USD/GBP	**0.9879**	0.4420	0.803	**0.9924**	0.2565	0.723	**0.9994**	0.0377	0.921	1.0039	-0.2067	0.900
JPY/USD	**0.9752**	1.4189	0.284	**0.9797**	1.8705	0.043 **	**0.9773**	1.2169	0.431	**0.9818**	1.9624	0.038 **
CHF/USD	**0.9735**	1.1157	0.490	**0.9921**	0.5904	0.586	**0.9627**	1.7485	0.220	**0.9811**	1.8243	0.084 *
CAD/USD	1.0037	-0.1624	0.946	1.0154	-0.9817	0.989	1.0009	-0.0372	0.948	1.0126	-0.7217	0.977
SEK/USD	**0.9724**	0.7632	0.649	**0.9850**	0.3936	0.626	**0.9698**	1.2796	0.402	**0.9823**	0.6637	0.561
DNK/USD	**0.9736**	0.9031	0.575	**0.9978**	0.1070	0.760	**0.9657**	0.9888	0.599	**0.9897**	0.7830	0.553
USD/AUD	1.0061	-0.3141	0.941	1.0250	-1.1795	0.980	**0.9972**	0.1147	0.815	1.0159	-0.7815	0.933
UIRP fundamental												
USD/GBP	**0.9864**	0.7234	0.765	**0.9781**	2.1231	0.025 **	**0.9857**	0.7515	0.791	**0.9774**	1.9200	0.060 *
JPY/USD	**0.9457**	0.9795	0.593	**0.9770**	1.1327	0.255	**0.9404**	1.0830	0.599	**0.9715**	1.3198	0.252
CHF/USD	**0.9783**	1.1936	0.448	**0.9908**	0.7151	0.469	**0.9764**	1.2342	0.450	**0.9889**	0.8070	0.498
CAD/USD	**0.9886**	0.8990	0.645	**0.9968**	0.3981	0.682	**0.9832**	1.3722	0.403	**0.9913**	1.1748	0.325
SEK/USD	**0.9739**	1.5075	0.294	**0.9728**	1.1926	0.248	**0.9714**	1.3769	0.403	**0.9702**	1.2760	0.282
DNK/USD	**0.9701**	1.1746	0.528	**0.9717**	1.2501	0.258	**0.9613**	1.2202	0.576	**0.9629**	1.2633	0.323
USD/AUD	**0.9668**	1.6115	0.232	**0.9752**	1.1766	0.289	**0.9596**	1.8529	0.147	**0.9680**	1.2606	0.262
Monetary model fundamentals												
USD/GBP	**0.8629**	1.2891	0.762	**0.9439**	1.5664	0.273	**0.8703**	1.2844	0.734	**0.9520**	1.7607	0.159
JPY/USD	**0.9418**	1.1146	0.776	**0.9988**	0.0490	0.912	**0.9303**	1.2984	0.648	**0.9866**	0.5495	0.745
CHF/USD	**0.9449**	1.3199	0.660	**0.9882**	0.9363	0.516	**0.9530**	1.2179	0.685	**0.9966**	0.2779	0.827
CAD/USD	**0.9243**	1.7267	0.479	**0.9906**	0.3506	0.855	**0.9112**	1.6916	0.458	**0.9766**	1.3857	0.286
SEK/USD	**0.9064**	1.8967	0.373	**0.9826**	0.5045	0.817	**0.8766**	1.6312	0.480	**0.9504**	1.3113	0.335
DNK/USD	**0.9385**	1.6310	0.363	**0.9879**	0.7062	0.550	**0.9431**	1.3435	0.529	**0.9927**	0.2514	0.773
USD/AUD	**0.9088**	2.1351	0.148	**0.9746**	1.0401	0.404	**0.8957**	1.9977	0.188	**0.9605**	1.3824	0.217
All fundamentals												
USD/GBP	**0.8957**	1.5646	0.915	**0.9683**	0.6492	0.919	**0.9011**	1.4781	0.930	**0.9741**	0.5058	0.951
JPY/USD	**0.8492**	2.0908	0.795	**0.9244**	1.5692	0.515	**0.8421**	2.1497	0.760	**0.9167**	1.6617	0.418
CHF/USD	**0.7509**	1.8858	0.759	**0.8826**	2.6233	0.089 *	**0.7549**	1.8643	0.765	**0.8873**	2.5946	0.086 *
CAD/USD	**0.8908**	2.4992	0.533	**0.9694**	1.0198	0.787	**0.8777**	2.4182	0.578	**0.9552**	1.4765	0.553
SEK/USD	**0.8813**	1.5078	0.925	**0.9260**	1.5598	0.575	**0.8565**	1.5039	0.927	**0.8999**	2.0743	0.255
DNK/USD	**0.8673**	2.6383	0.550	**0.9094**	2.3560	0.196	**0.8638**	3.1635	0.268	**0.9058**	1.7617	0.564
USD/AUD	**0.8228**	2.2132	0.643	**0.9417**	1.7659	0.386	**0.8126**	2.2328	0.624	**0.9300**	2.1663	0.168

Notes: This table presents the comparison of forecasting performance between, respectively, SRidge and EWA vs. the rolling and recursive OLS for decoupled fundamentals for the period February 1999–December 2014 using real-time data. Columns 1-6 show the comparison between SRidge and OLS methods, while columns 7-12 for EWA vs. OLS. For each comparison, the Theil ratio (RMSE of the given method / RMSE of the compared) and the DM statistic and its bootstrapped p-value are shown. The DM test is a one-sided test of equal out-of-sample prediction accuracy (H_0) against superior out-of-sample prediction accuracy (H_1) of the machine-learning method considered against the OLS methods. ***, **, and * denote statistical significance at the 1 %, 5 %, and 10 % levels. Theil ratios < 1 are indicated in bold.

Table D.23: Directional predictions of the exchange rates based on decoupled fundamentals, for a sample February 1999 – December 2014.

Currency pair	Rolling regression			Recursive regression			Sridge			EWA		
	Proportion of changes predicted	DM statistic	DM p-value	Proportion of changes predicted	DM statistic	DM p-value	Proportion of changes predicted	DM statistic	DM p-value	Proportion of changes predicted	DM statistic	DM p-value
PPP fundamental												
USD/GBP	**0.523**	0.5161	0.233	**0.546**	0.9781	0.117	**0.531**	0.5966	0.232	0.423	-1.3965	0.914
JPY/USD	**0.515**	0.3125	0.376	0.431	-1.4799	0.954	**0.508**	0.1475	0.452	**0.538**	0.8166	0.166
CHF/USD	**0.500**	0.0000	0.428	0.462	-0.7667	0.769	**0.500**	0.0000	0.427	0.477	-0.4022	0.632
CAD/USD	0.485	-0.2480	0.620	0.462	-0.8390	0.822	0.454	-1.0118	0.867	**0.515**	0.3437	0.370
SEK/USD	**0.585**	1.8765	0.017 **	**0.592**	1.8792	0.018 **	**0.546**	1.0091	0.118	0.462	-0.6648	0.768
DNK/USD	**0.606**	1.8115	0.032 **	**0.535**	0.6090	0.274	**0.535**	0.5752	0.308	0.472	-0.5135	0.812
USD/AUD	**0.532**	0.5249	0.479	**0.532**	0.4497	0.565	**0.508**	0.1367	0.707	0.444	-0.9448	0.814
UIRP fundamental												
USD/GBP	0.485	-0.3510	0.629	0.469	-0.6804	0.737	**0.538**	0.8797	0.149	**0.523**	0.4562	0.317
JPY/USD	0.492	-0.0993	0.540	**0.508**	0.1336	0.465	**0.562**	0.9645	0.147	0.492	-0.1405	0.588
CHF/USD	**0.515**	0.3184	0.317	0.492	-0.1531	0.472	0.492	-0.1505	0.472	0.485	-0.2326	0.584
CAD/USD	0.469	-0.6725	0.795	**0.500**	0.0000	0.500	**0.515**	0.3233	0.411	**0.515**	0.3149	0.384
SEK/USD	**0.519**	0.3759	0.346	0.411	-1.7438	0.978	**0.512**	0.2140	0.431	**0.512**	0.2362	0.406
DNK/USD	0.496	-0.0616	0.516	0.472	-0.4992	0.715	**0.528**	0.4657	0.299	**0.559**	1.0493	0.134
USD/AUD	0.408	-1.4826	0.938	0.492	-0.1739	0.606	0.477	-0.4235	0.697	0.462	-0.7818	0.727
Monetary model fundamentals												
USD/GBP	0.485	-0.2788	0.656	0.446	-1.1149	0.882	**0.508**	0.1704	0.434	0.492	-0.1593	0.603
JPY/USD	**0.550**	1.0004	0.157	**0.574**	1.6913	0.039 **	**0.566**	1.5099	0.064 *	**0.574**	1.6913	0.035 **
CHF/USD	**0.524**	0.4866	0.349	0.468	-0.5743	0.822	**0.508**	0.1624	0.499	0.468	-0.6499	0.815
CAD/USD	**0.550**	1.0172	0.183	**0.543**	0.9584	0.220	**0.519**	0.4242	0.437	0.488	-0.2642	0.777
SEK/USD	0.457	-0.8225	0.844	0.450	-1.0794	0.907	0.411	-1.8715	0.983	0.434	-1.2723	0.961
DNK/USD	**0.534**	0.7186	0.370	**0.534**	0.6008	0.474	**0.500**	0.0000	0.677	0.483	-0.3644	0.775
USD/AUD	**0.508**	0.1714	0.560	0.475	-0.4022	0.789	0.458	-0.7747	0.893	0.458	-0.8710	0.969
All fundamentals												
USD/GBP	**0.515**	0.2212	0.481	**0.554**	1.2351	0.125	**0.508**	0.1571	0.442	0.462	-0.7906	0.815
JPY/USD	**0.620**	2.7917	0.008 ***	0.496	-0.0748	0.612	**0.566**	1.5099	0.053 *	**0.566**	1.0851	0.129
CHF/USD	**0.516**	0.3257	0.373	0.468	-0.7199	0.817	**0.500**	0.0000	0.560	0.444	-0.8123	0.894
CAD/USD	**0.543**	0.9720	0.247	0.496	-0.0874	0.760	**0.519**	0.4370	0.455	**0.504**	0.0880	0.608
SEK/USD	**0.589**	1.5226	0.054 *	**0.550**	1.0794	0.162	0.442	-1.1727	0.932	0.457	-0.7505	0.886
DNK/USD	0.492	-0.1179	0.650	**0.585**	1.3433	0.155	**0.508**	0.1741	0.635	**0.517**	0.3684	0.539
USD/AUD	0.475	-0.4563	0.824	0.483	-0.3108	0.787	0.475	-0.5312	0.861	0.442	-1.0788	0.976

Notes: This table presents the results of forecasting the direction of change of the end-of-month exchange rate 1-month ahead by the rolling OLS, recursive OLS, SRidge and EWA methods for a sample February 1999 – December 2014, based on decoupled fundamentals. For each method, the share of correctly forecasted changes, the DM statistic and its bootstrapped p-value are shown. The DM tests is a one-sided test of equal out-of-sample prediction accuracy (H_0) against superior out-of-sample prediction accuracy (H_1) for the methods considered against the benchmark of a 50 % success rate. ***, **, and * denote statistical significance at the 1 %, 5 %, and 10 % levels. Shares > 0.5 are indicated in bold.

Table D.24: Directional predictions of the exchange rates on the real-time data set (February 1999 – December 2014) based on decoupled fundamentals.

Currency pair	Rolling regression: Proportion of changes predicted	Rolling regression: DM statistic	Rolling regression: DM p-value	Recursive regression: Proportion of changes predicted	Recursive regression: DM statistic	Recursive regression: DM p-value	Sridge: Proportion of changes predicted	Sridge: DM statistic	Sridge: DM p-value	EWA: Proportion of changes predicted	EWA: DM statistic	EWA: DM p-value
PPP fundamental												
USD/GBP	**0.546**	1.0570	0.128	**0.569**	1.5941	0.041 **	**0.523**	0.4883	0.301	0.431	-1.4638	0.921
JPY/USD	**0.500**	0.0000	0.472	0.477	-0.4804	0.718	**0.515**	0.3113	0.373	**0.531**	0.6079	0.254
CHF/USD	**0.500**	0.0000	0.408	0.469	-0.4707	0.649	0.454	-0.9066	0.807	0.477	-0.3574	0.504
CAD/USD	**0.523**	0.4535	0.341	0.477	-0.4268	0.729	0.485	-0.3251	0.683	**0.515**	0.3510	0.375
SEK/USD	**0.585**	1.7247	0.051 *	**0.569**	1.3036	0.089 *	**0.531**	0.6690	0.264	0.469	-0.5783	0.696
DNK/USD	**0.543**	0.9696	0.149	**0.528**	0.4753	0.334	**0.528**	0.4657	0.339	0.480	-0.3650	0.723
USD/AUD	**0.500**	0.0000	0.660	**0.532**	0.4800	0.585	**0.524**	0.4950	0.585	0.387	-2.0931	0.947
UIRP fundamental												
USD/GBP	0.485	-0.3510	0.624	0.469	-0.6804	0.730	**0.538**	0.8797	0.159	**0.523**	0.4562	0.305
JPY/USD	0.492	-0.0993	0.515	**0.508**	0.1336	0.458	**0.562**	0.9645	0.152	0.492	-0.1405	0.542
CHF/USD	**0.515**	0.3184	0.309	0.492	-0.1531	0.523	0.492	-0.1505	0.500	0.485	-0.2326	0.577
CAD/USD	0.469	-0.6725	0.811	**0.500**	0.0000	0.544	**0.515**	0.3233	0.437	**0.515**	0.3149	0.414
SEK/USD	**0.519**	0.3759	0.372	0.411	-1.7438	0.974	**0.512**	0.2140	0.430	**0.512**	0.2362	0.391
DNK/USD	0.496	-0.0616	0.493	0.472	-0.4992	0.689	**0.528**	0.4657	0.259	**0.559**	1.0493	0.118
USD/AUD	0.408	-1.4826	0.963	0.492	-0.1739	0.598	0.477	-0.4235	0.722	0.462	-0.7818	0.729
Monetary model fundamentals												
USD/GBP	0.469	-0.6489	0.743	0.400	-1.9319	0.976	**0.515**	0.3417	0.335	0.469	-0.6692	0.740
JPY/USD	**0.543**	0.9720	0.130	**0.612**	2.3688	0.007 ***	**0.558**	1.2512	0.079 *	**0.550**	1.0169	0.104
CHF/USD	0.484	-0.3554	0.622	0.435	-1.0145	0.843	0.476	-0.3926	0.669	0.403	-2.0581	0.985
CAD/USD	0.457	-0.8893	0.880	0.426	-1.6913	0.974	0.473	-0.5648	0.816	0.457	-0.7214	0.880
SEK/USD	**0.504**	0.0755	0.500	0.450	-0.7721	0.810	**0.504**	0.0698	0.477	**0.543**	0.9320	0.118
DNK/USD	**0.534**	0.6254	0.352	**0.517**	0.2973	0.565	**0.559**	1.0093	0.226	**0.500**	0.0000	0.592
USD/AUD	**0.517**	0.3654	0.398	**0.500**	0.0000	0.571	0.475	-0.4767	0.743	0.442	-1.1605	0.850
All fundamentals												
USD/GBP	**0.515**	0.3510	0.351	**0.615**	2.7042	0.003 ***	**0.515**	0.3417	0.338	**0.508**	0.1441	0.425
JPY/USD	**0.504**	0.0686	0.572	**0.535**	0.6551	0.341	**0.550**	1.0366	0.123	**0.543**	0.9225	0.138
CHF/USD	**0.532**	0.6475	0.247	0.484	-0.2589	0.614	**0.516**	0.2805	0.380	0.403	-2.0581	0.973
CAD/USD	0.481	-0.3087	0.775	**0.512**	0.2595	0.594	0.465	-0.7711	0.861	0.434	-1.4604	0.953
SEK/USD	0.465	-0.7860	0.847	0.450	-1.1444	0.923	**0.527**	0.5009	0.310	**0.558**	1.3279	0.052 *
DNK/USD	**0.500**	0.0000	0.568	0.492	-0.1416	0.643	**0.559**	0.9985	0.261	**0.508**	0.1122	0.633
USD/AUD	**0.508**	0.1826	0.580	**0.550**	0.7689	0.354	0.475	-0.4767	0.872	0.450	-1.0493	0.852

Notes: This table presents the results of forecasting the direction of change of the end-of-month exchange rate 1-month ahead by the rolling OLS, recursive OLS, SRidge and EWA methods based on decoupled fundamentals for the period February 1999–December 2014 using real-time data. For each method, the share of correctly forecasted changes, the DM statistic and its bootstrapped p-value are shown. The DM tests is a one-sided test of equal out-of-sample prediction accuracy (H_0) against superior out-of-sample prediction accuracy (H_1) for the methods considered against the benchmark of a 50 % success rate. ***, **, and * denote statistical significance at the 1 %, 5 %, and 10 % levels. Shares > 0.5 are indicated in bold.

Table D.25: 1-month ahead forecasts using decoupled Taylor-rule fundamentals.

Currency pair	No change	Rolling regression				Recursive regression				SRidge				EWA			
	RMSE x 100	Theil ratio	CW p-value	DM statistic	DM p-value	Theil ratio	CW p-value	DM statistic	DM p-value	Theil ratio	CW p-value	DM statistic	DM p-value	Theil ratio	CW p-value	DM statistic	DM p-value
Output gap fundamentals: deviations from a linear trend																	
USD/GBP	2.9051	1.0430	0.192	-2.1392	0.211	1.0276	0.566	-1.6732	0.489	1.0011	0.493	-0.2706	0.237	1.0030	0.185	-0.3110	0.461
JPY/USD	3.1046	1.0297	0.098 *	-1.3703	0.061 *	1.0048	0.025 **	-0.3162	0.060 *	**0.9996**	0.204	0.7441	0.090 *	**0.9937**	0.016 **	0.8878	0.077 *
CHF/USD	3.3887	1.0644	0.535	-2.7086	0.639	1.0216	0.508	-1.5669	0.553	**0.9986**	0.097 *	1.0748	0.056 *	**0.9941**	0.028 **	1.0664	0.078 *
CAD/USD	1.9994	1.0145	0.287	-0.7791	0.005 ***	**0.9988**	0.137	0.1123	0.006 ***	1.0002	0.640	-0.3738	0.303	**0.9995**	0.201	0.4814	0.161
SEK/USD	3.2263	1.1008	0.837	-1.8130	0.503	1.0241	0.650	-2.3986	0.927	**0.9971**	0.097 *	0.2861	0.190	**0.9954**	0.012 **	0.8034	0.060 *
DNK/USD	3.1037	1.0323	0.265	-1.5297	0.092 *	1.0136	0.361	-1.2282	0.305	1.0023	0.895	-1.2469	0.792	**0.9939**	0.028 **	0.7766	0.137
USD/AUD	3.3926	1.0073	0.007 ***	-0.3638	0.002 ***	1.0126	0.533	-1.0460	0.198	1.0043	0.752	-0.9165	0.612	1.0009	0.375	-0.1571	0.346
FRF/USD	3.1978	1.0359	0.430	-1.3733	0.196	1.0277	0.510	-1.3370	0.429	1.0005	0.443	-0.0983	0.205	**0.9961**	0.085 *	0.3001	0.212
DEM/USD	3.3136	1.0444	0.658	-1.5401	0.314	1.0303	0.612	-1.4812	0.572	**0.9987**	0.213	0.5630	0.093 *	**0.9914**	0.042 **	0.8058	0.053 *
ITL/USD	3.1907	1.0082	0.116	-0.3347	0.031 **	1.0216	0.814	-1.5907	0.705	1.0024	0.339	-0.1958	0.547	1.0067	0.276	-0.4274	0.780
NLG/USD	3.3319	1.0341	0.256	-1.1281	0.152	1.0212	0.551	-1.2831	0.422	**0.9998**	0.394	0.1412	0.129	**0.9883**	0.022 **	1.0668	0.030 **
PTE/USD	2.7703	1.0606	0.714	-1.3914	0.604	1.0976	0.724	-1.9837	0.902	**0.9970**	0.365	0.2264	0.177	**0.9987**	0.338	0.1858	0.320
Output gap fundamentals: deviations from a quadratic trend																	
USD/GBP	2.9051	1.0387	0.157	-1.9223	0.144	1.0298	0.681	-1.9623	0.664	1.0010	0.482	-0.2455	0.250	1.0027	0.155	-0.2798	0.499
JPY/USD	3.1046	1.0238	0.056 *	-1.1279	0.032 **	1.0063	0.048 **	-0.4245	0.065 *	**0.9995**	0.156	0.9260	0.060 *	**0.9943**	0.022 **	0.7911	0.136
CHF/USD	3.3887	1.0601	0.333	-2.2897	0.402	1.0223	0.713	-1.9219	0.727	**0.9986**	0.099 *	1.0649	0.028 **	**0.9927**	0.016 **	1.3013	0.053 *
CAD/USD	1.9994	1.0184	0.246	-0.8391	0.005 ***	**0.9991**	0.172	0.0883	0.004 ***	1.0002	0.636	-0.3663	0.308	**0.9993**	0.136	0.7874	0.086 *
SEK/USD	3.2263	1.0756	0.757	-1.5823	0.438	1.0314	0.853	-2.1039	0.849	**0.9986**	0.132	0.1385	0.243	**0.9951**	0.010 **	0.8495	0.032 **
DNK/USD	3.1037	1.0270	0.152	-1.1256	0.032 **	1.0154	0.723	-1.8647	0.647	1.0005	0.892	-1.0250	0.660	**0.9933**	0.024 **	0.8476	0.119
USD/AUD	3.3926	1.0063	0.003 ***	-0.2918	0.000 ***	1.0096	0.242	-0.7776	0.081 *	1.0043	0.747	-0.9038	0.600	1.0012	0.429	-0.2275	0.419
FRF/USD	3.1978	1.0333	0.406	-1.3364	0.199	1.0279	0.566	-1.3693	0.468	1.0005	0.444	-0.0991	0.212	**0.9964**	0.087 *	0.2784	0.206
DEM/USD	3.3136	1.0456	0.674	-1.6310	0.387	1.0219	0.599	-1.2336	0.457	**0.9987**	0.213	0.5639	0.101	**0.9908**	0.037 **	0.8704	0.059 *
ITL/USD	3.1907	1.0029	0.083 *	-0.1237	0.020 **	1.0206	0.851	-1.5380	0.714	1.0024	0.339	-0.1960	0.561	1.0047	0.238	-0.3022	0.696
NLG/USD	3.3319	1.0356	0.272	-1.1889	0.211	1.0146	0.676	-1.2050	0.468	**0.9998**	0.395	0.1386	0.157	**0.9883**	0.022 **	1.0623	0.042 **
PTE/USD	2.7703	1.0501	0.605	-0.9218	0.397	1.0848	0.738	-1.7149	0.848	**0.9970**	0.366	0.2209	0.172	**0.9988**	0.351	0.1784	0.278
Output gap fundamentals: deviations from a linear and a quadratic trend																	
USD/GBP	2.9051	1.0353	0.254	-1.8864	0.151	1.0262	0.846	-2.2301	0.805	1.0010	0.488	-0.2580	0.238	1.0014	0.120	-0.1427	0.418
JPY/USD	3.1046	1.0301	0.081 *	-1.5177	0.079 *	1.0042	0.044 **	-0.3260	0.061 *	**0.9996**	0.207	0.7351	0.085 *	**0.9943**	0.021 **	0.8010	0.069 *
CHF/USD	3.3887	1.0461	0.257	-1.8590	0.209	1.0178	0.557	-1.4286	0.492	**0.9986**	0.099 *	1.0619	0.044 **	**0.9930**	0.018 **	1.2581	0.058 *
CAD/USD	1.9994	1.0167	0.211	-0.7619	0.002 ***	**0.9982**	0.177	0.1767	0.004 ***	1.0002	0.643	-0.3792	0.343	**0.9990**	0.074 *	0.9824	0.060 *
SEK/USD	3.2263	1.0639	0.837	-2.2567	0.663	1.0272	0.868	-2.3581	0.923	**0.9982**	0.125	0.1789	0.230	**0.9958**	0.014 **	0.7365	0.060 *
DNK/USD	3.1037	1.0298	0.167	-1.2546	0.050 **	1.0136	0.627	-1.5159	0.490	1.0004	0.882	-0.9723	0.665	**0.9926**	0.020 **	0.9363	0.119
USD/AUD	3.3926	1.0029	0.003 ***	-0.1332	0.000 ***	1.0052	0.181	-0.4199	0.038 **	1.0042	0.741	-0.8893	0.602	1.0021	0.472	-0.3543	0.463
FRF/USD	3.1978	1.0363	0.243	-1.2355	0.152	1.0263	0.570	-1.3011	0.415	1.0005	0.443	-0.0983	0.202	**0.9951**	0.073 *	0.3733	0.174
DEM/USD	3.3136	1.0431	0.665	-1.5888	0.398	1.0174	0.371	-0.8289	0.270	**0.9987**	0.213	0.5633	0.074 *	**0.9905**	0.035 **	0.8967	0.054 *
ITL/USD	3.1907	1.0160	0.118	-0.5766	0.084 *	1.0182	0.732	-1.1898	0.571	1.0025	0.344	-0.2033	0.609	1.0075	0.294	-0.4786	0.818
NLG/USD	3.3319	1.0352	0.281	-1.1757	0.183	1.0171	0.618	-1.0907	0.346	**0.9998**	0.395	0.1382	0.158	**0.9889**	0.025 **	1.0125	0.048 **
PTE/USD	2.7703	1.0489	0.587	-0.9945	0.424	1.0811	0.727	-1.7652	0.848	**0.9970**	0.365	0.2264	0.175	**0.9987**	0.339	0.1862	0.300
Output gap fundamentals: deviations from a Hodrick-Prescott filtered trend																	
USD/GBP	2.9051	1.0427	0.555	-2.2624	0.372	1.0203	0.750	-1.6556	0.604	1.0011	0.495	-0.2758	0.274	1.0026	0.153	-0.2692	0.466
JPY/USD	3.1046	1.0233	0.150	-1.2881	0.064 *	1.0034	0.015 **	-0.2266	0.060 *	**0.9996**	0.175	0.8554	0.066 *	**0.9944**	0.022 **	0.7746	0.064 *
CHF/USD	3.3887	1.0504	0.673	-2.9215	0.796	1.0220	0.757	-1.7578	0.704	**0.9986**	0.094 *	1.0916	0.037 **	**0.9938**	0.025 **	1.1083	0.074 *
CAD/USD	1.9994	1.0167	0.274	-1.2248	0.019 **	1.0041	0.650	-0.9388	0.140	1.0002	0.643	-0.3797	0.322	**0.9992**	0.078 *	0.9625	0.060 *
SEK/USD	3.2263	1.0679	0.806	-2.2239	0.671	1.0300	0.841	-2.0696	0.911	**0.9972**	0.099 *	0.2832	0.214	**0.9954**	0.012 **	0.8077	0.037 **
DNK/USD	3.1037	1.0322	0.343	-2.0045	0.345	1.0086	0.447	-0.8599	0.212	1.0004	0.881	-0.9637	0.673	**0.9928**	0.022 **	0.9097	0.113
USD/AUD	3.3926	1.0041	0.007 ***	-0.1774	0.000 ***	1.0121	0.520	-0.9978	0.203	1.0043	0.747	-0.9041	0.603	1.0019	0.468	-0.3408	0.459
FRF/USD	3.1978	1.0361	0.374	-1.4169	0.237	1.0283	0.638	-1.3603	0.478	1.0005	0.444	-0.0987	0.221	**0.9951**	0.072 *	0.3738	0.180
DEM/USD	3.3136	1.0482	0.835	-2.0596	0.685	1.0239	0.704	-1.2549	0.508	**0.9987**	0.214	0.5635	0.093 *	**0.9913**	0.041 **	0.8159	0.058 *
ITL/USD	3.1907	1.0190	0.117	-0.6804	0.115	1.0218	0.844	-1.4435	0.682	1.0024	0.340	-0.1977	0.589	1.0077	0.296	-0.4864	0.797
NLG/USD	3.3319	1.0381	0.393	-1.3368	0.294	1.0174	0.754	-1.2549	0.488	**0.9998**	0.396	0.1380	0.155	**0.9889**	0.025 **	1.0135	0.048 **
PTE/USD	2.7703	1.0405	0.365	-0.8804	0.366	1.0530	0.736	-1.7849	0.850	**0.9970**	0.365	0.2271	0.180	**0.9987**	0.343	0.1779	0.292

Notes: This table presents the results of forecasting end-of-month exchange rate 1-month ahead by the no-change prediction, the rolling OLS, recursive OLS, SRidge and EWA methods for different sets of Taylor-rule fundamentals. Column 1 shows the RMSE values for the no-change prediction. For each method, the Theil ratio (RMSE of the given method / RMSE of the no-change prediction), the CW-p-values, the DM statistic and its bootstrapped p-value are shown. The CW and DM tests are one-sided tests of equal out-of-sample prediction accuracy (H_0) against superior out-of-sample prediction accuracy (H_1) of the methods considered compared to the no-change prediction. ***, **, and * denote statistical significance at the 1 %, 5 %, and 10 % levels. Theil ratios < 1 are indicated in bold. Sample period for the exchange rates: March 1973 – December 2014 unless shorter (indicated in Appendix A).

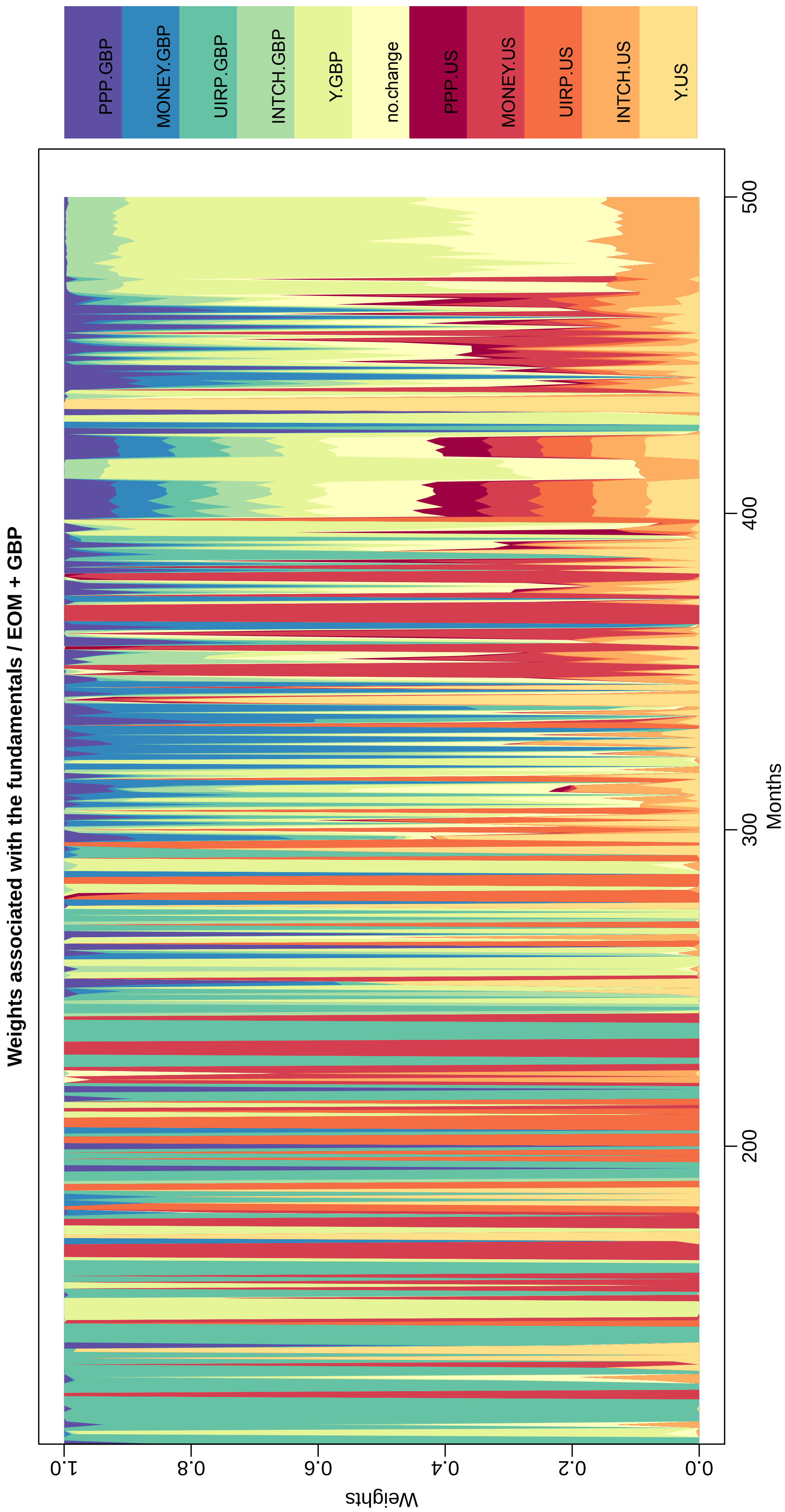

Figure D.1: Weights of different fundamentals in prediction: the USD/GBP case, all fundamentals considered together.

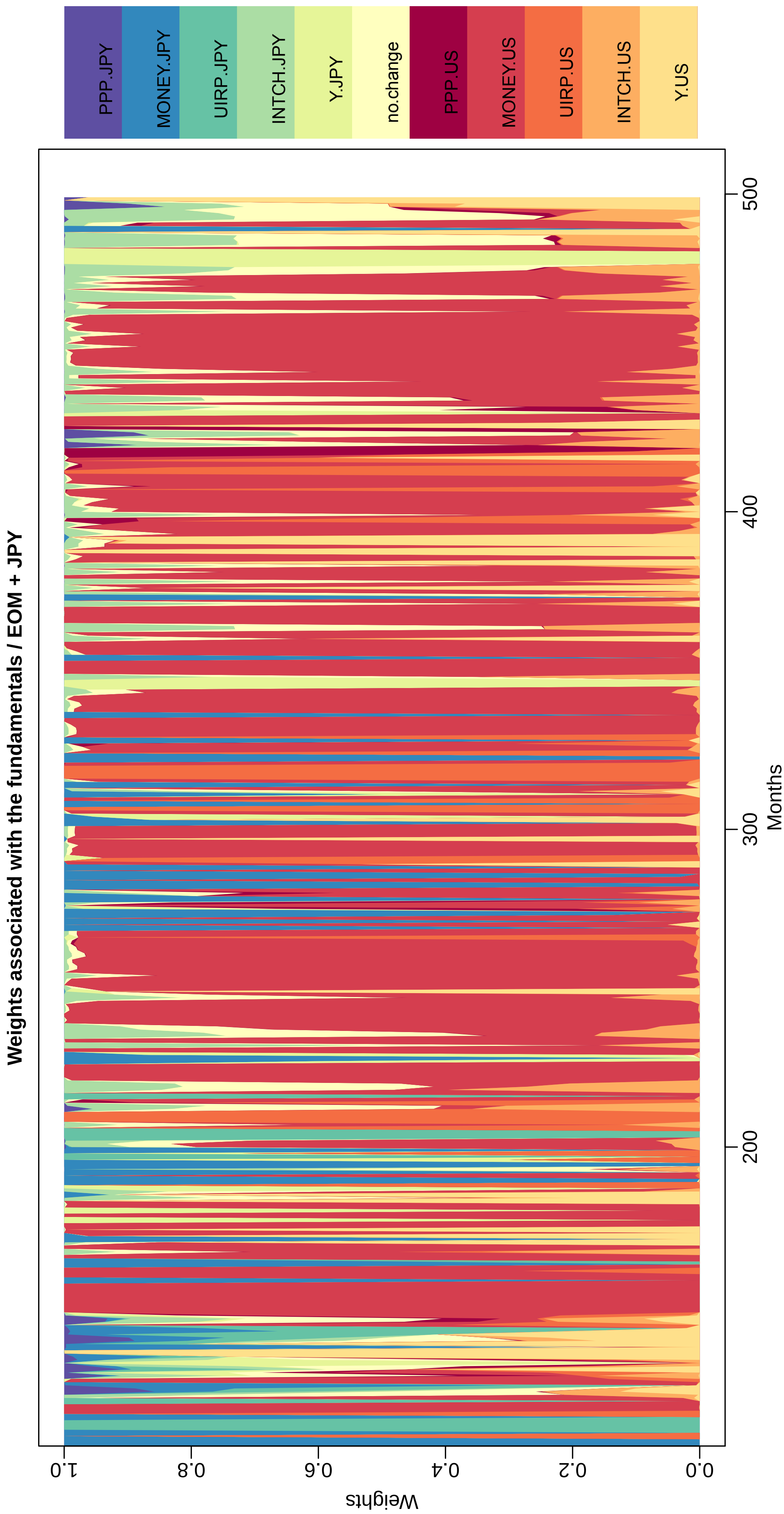

Figure D.2: Weights of different fundamentals in prediction: the JPY/USD case, all fundamentals considered together.

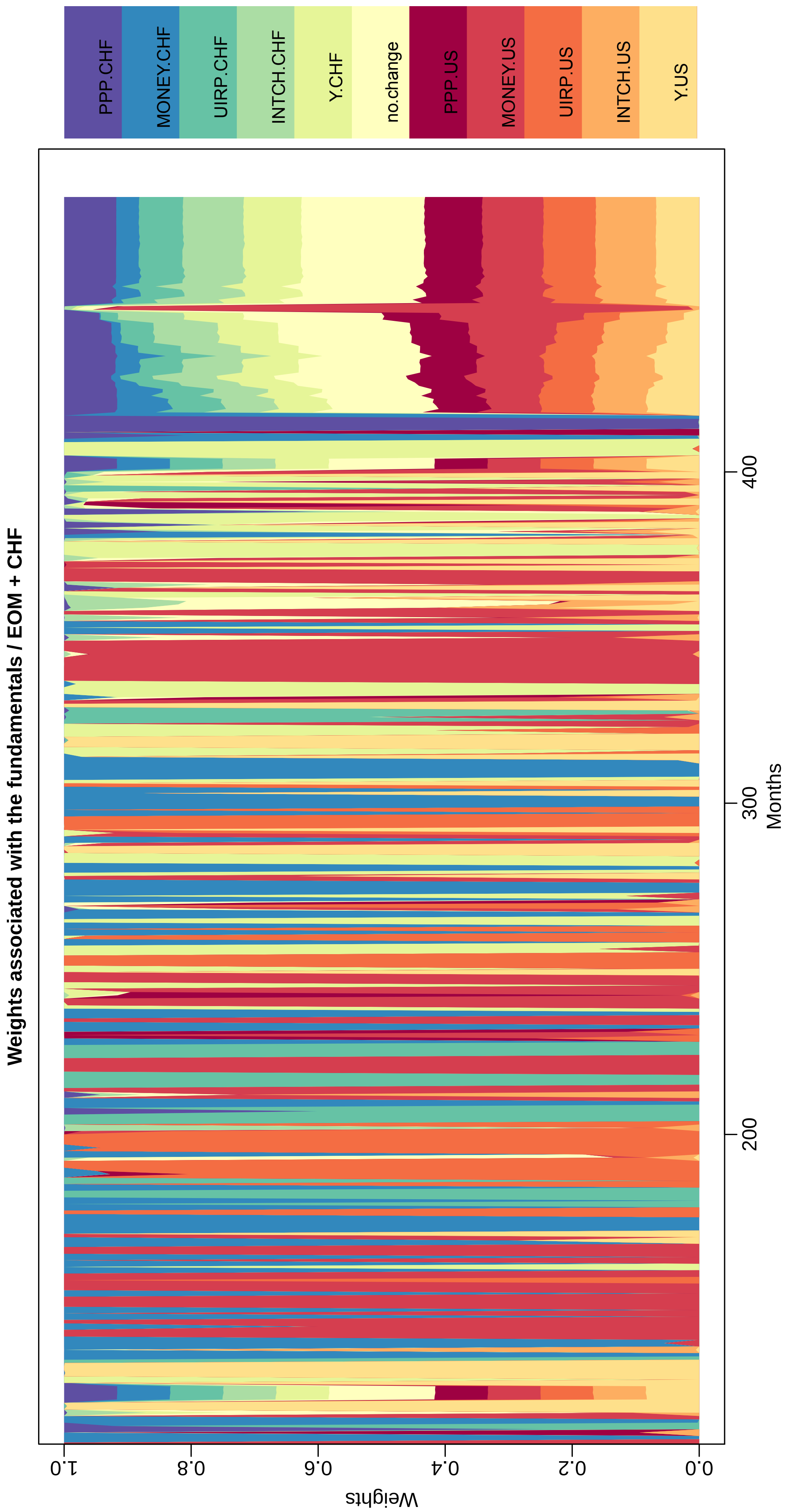

Figure D.3: Weights of different fundamentals in prediction: the CHF/USD case, all fundamentals considered together.